

Stable, crystalline, porous, covalent organic frameworks as a platform for chiral organocatalysts

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Supplementary Tables**Supplementary Table 1** | Calculated crystal stacking energy for TPB-TP-COF, TPB-DHTP-COF and TPB-DMTP-COF

COFs	E_{stack}	E_{L}	E_{e}
TPB-TP-COF	94.084	97.632	3.548
TPB-DHTP-COF	102.597	102.639	0.042
TPB-DMTP-COF	106.862	107.129	0.267

E_{stack} = crystal stacking energy, E_{L} = London dispersion energy, E_{e} = electronic energy. Energies are per unit cell and are given in kcal mol⁻¹.

Supplementary Table 2 | Full width at half maximum (FWHM) of the (100) peak at 2.76° in the PXRD patterns of TPB-DMTP-COF and [S-Py]_{0.17}-TPB-DMTP-COF

TPB-DMTP-COF	FWHM	[S-Py] _{0.17} -TPB-DMTP-COF	FWHM
As synthesized	0.39	As synthesized	0.437
DMF for 1 week	0.406	DMF for 1 week	0.391
DMSO for 1 week	0.403	DMSO for 1 week	0.449
THF for 1 week	0.402	THF for 1 week	0.437
MeOH for 1 week	0.411	MeOH for 1 week	0.406
Cyclohexanone for 1 week	0.404	Cyclohexanone for 1 week	0.417
12 M HCl for 1 week	0.468	12 M HCl for 1 week	0.441
Water (25 °C) for 1 week	0.401	Water (25 °C) for 1 week	0.404
Water (100 °C) for 1 week	0.46	Water (100 °C) for 1 week	0.465
14 M NaOH for 1 week	0.379	14 M NaOH for 1 week	0.454

Supplementary Table 3 | Surface area, pore size and pore volume of COFs

COFs	BET Surface Area (m ² g ⁻¹)	Langmuir Surface Area (m ² g ⁻¹)	Pore Size (nm)	Pore Volume (cm ³ g ⁻¹)
TPB-DMTP-COF	2105	3336	3.26	1.28
TPB-DMTP-COF (HCl, 12 M, 1 week)	2074	3299	3.26	1.23
TPB-DMTP-COF (H ₂ O, 100 °C, 1 week)	2081	3321	3.25	1.27
TPB-DMTP-COF (NaOH, 14 M, 1 week)	2020	3126	3.25	1.18
[HC≡C] _{0.17} -TPB-DMTP-COF	1973	3131	3.17	1.19
[HC≡C] _{0.34} -TPB-DMTP-COF	1760	2791	3.1	1.12
[HC≡C] _{0.50} -TPB-DMTP-COF	1642	2552	3.03	1.02
[HC≡C] _{1.0} -TPB-DMTP-COF	1297	1765	2.75	0.58
[S-Py] _{0.17} -TPB-DMTP-COF	1970	3059	3.07	1.13
[S-Py] _{0.17} -TPB-DMTP-COF (Five cycled)	1916	2818	2.98	1.08
[S-Py] _{0.34} -TPB-DMTP-COF	1802	2812	2.95	1.04
[S-Py] _{0.34} -TPB-DMTP-COF	1612	2300	2.86	0.83
[S-Py] _{0.17} -TPB-DMTP-COF (HCl, 12 M, 1 week)	1956	3049	3.07	1.12
[S-Py] _{0.17} -TPB-DMTP-COF (NaOH, 14 M, 1 week)	1971	3050	3.07	1.14
[S-Py] _{0.17} -TPB-DMTP-COF (H ₂ O, 100 °C, 1 week)	1924	2943	3.07	1.09
TPB-DHTP-COF	849	1256	Multiple pore	0.44
TPB-DHTP-COF (HCl, 12 M, 1 week)	84	123	Multiple pore	0.047
TPB-DHTP-COF (H ₂ O, 100 °C, 1 week)	264	385	Multiple pore	0.14
TPB-DHTP-COF (NaOH, 14 M, 1 week) ^a	-	-	-	-
TPB-TP-COF	16	22	-	-

^anonporous

Supplementary Table 4 | Stability of different COFs reported to date

Name	Conditions	PXRD	BET Surface Area	Reference
TPB-DMTP-COF	12 M HCl (7 days)	Retained intensities and peak positions	Pristine: 2105 m ² g ⁻¹ ; 7 days later: 2074 m ² g ⁻¹	This work
	Boiling water (7 days)	Retained intensities and peak positions	Pristine: 2105 m ² g ⁻¹ ; 7days later: 2081 m ² g ⁻¹	
	14 M NaOH (7 days)	Retained intensities and peak positions	Pristine: 2105 m ² g ⁻¹ ; 7days later: 2020 m ² g ⁻¹	
TpPa-1	Boiling water (7 days)	Retained intensities and peak positions	Pristine: 535 m ² g ⁻¹ ; 7days later: 520 m ² g ⁻¹	<i>J. Am. Chem. Soc.</i> 134 , 19524-19527, (2012).
	9 M HCl (7 days)	Retained intensities and peak positions	Pristine: 535 m ² g ⁻¹ ; 7days later: 512 m ² g ⁻¹	
	9 M NaOH (1 day)	Loss peaks and intensities	Pristine: 535 m ² g ⁻¹ ; 7days later: 324 m ² g ⁻¹	
Py-Azine COF	1M NaOH (1 day); 1M HCl (1 day); H ₂ O (1 day); MeOH (1 day); Hexane (1 day).	Retained intensities and peak positions	Pristine: 1210 m ² g ⁻¹	<i>J. Am. Chem. Soc.</i> 135 , 17310-17313, (2013).
TpBD TpBD(MC)	Boiling water (7 days)	Loss peaks and intensities	Pristine: 537 m ² g ⁻¹ ; 7 days later: 35 m ² g ⁻¹	<i>J. Am. Chem. Soc.</i> 135 , 5328-5331, (2013).
TpPa-2 TpPa-2(MC)	9 M HCl (7 days)	Loss peaks and intensities	Pristine: 339 m ² g ⁻¹ ; 7 days later: 56 m ² g ⁻¹	

TpPa-2 TpPa-2(MC)	9 M NaOH (7 days)	Loss peaks and intensities	Pristine: 339 m ² g ⁻¹ ; 7 days later: 56 m ² g ⁻¹	
DhaTph	3 M HCl (7 days) Boiling water (7 days)	Loss intensities	Pristine: 1305 m ² g ⁻¹ ; 7 days later: 570 m ² g ⁻¹ (3M HCl); 1252 m ² g ⁻¹ (Boiling water)	<i>Angew. Chem. Int. Ed.</i> 52 , 13052-13056, (2013).
	3 M NaOH (7 days)	Loss peaks and intensities	70% weight loss, the surface area information is not mentioned.	
DmaTph	3 M HCl (1 day) Boiling water (7 days)	Loss peaks and intensities	Pristine: 431 m ² g ⁻¹	
Tp-Azo	9 M HCl (7 days) Boiling water (7 days)	Loss intensities	Pristine: 1328 m ² g ⁻¹	<i>J. Am. Chem. Soc.</i> 136 , 6570-6573, (2014).
	1, 3 and 6 M NaOH (3 days)	Loss intensities	Pristine: 1328 m ² g ⁻¹	
Tp-Stb	9 M HCl (7 days)	Loss intensities	Pristine: 422 m ² g ⁻¹	
	3 and 6 M NaOH (3 days)	Loss intensities	Pristine: 422 m ² g ⁻¹	
CTV-COF-1	Water for 5 h, 10 h and 48 h.	Loss intensities	Pristine: 1245 m ² g ⁻¹ ; 48 h later: 997 m ² g ⁻¹	<i>Chem. Commun.</i> 50 , 788-791, (2014).
COF-10	Under 50% R.H atmosphere for 2.5 h, 5 h, 20 h and 4 days.	Loss intensities	Pristine: 1200 m ² g ⁻¹ ; 4 days later: 30 m ² g ⁻¹	<i>Chem. Commun.</i> 48 , 4606-4608, (2012).
Pyridine modified COF-10	Under 50% R.H atmosphere for 1.5 h, 20 h, 4 days and 7 days.	Retained intensities and peak positions	Pristine: 1200 m ² g ⁻¹ ; 7 days later: 950 m ² g ⁻¹	

Supplementary Table 5 | Atomistic coordinates for the AA-stacking mode of TPB-DMTP-COF optimized by using DFTB+ method (space group $P6$, $a = b = 37.2718 \text{ \AA}$, $c = 3.5215 \text{ \AA}$, $\alpha = \beta = 90^\circ$ and $\gamma = 120^\circ$)

Atom	x/a	y/b	z/c
C1	0.28981	0.64218	0.50948
C2	0.31459	0.62381	0.50941
C3	0.24366	0.61623	0.51129
C4	0.36807	0.58669	0.38149
C5	0.39171	0.56757	0.38659
C6	0.43264	0.5889	0.52061
C7	0.44902	0.63002	0.64034
C8	0.42488	0.64857	0.64092
N9	0.45939	0.5724	0.51985
C10	0.44488	0.53315	0.57564
C11	0.47293	0.51623	0.56578
C12	0.45731	0.47302	0.55581
C13	0.48409	0.45714	0.56395
O14	0.41445	0.44831	0.5392
C15	0.6025	0.59337	0.4059
H16	0.29983	0.59004	0.50963
H17	0.33674	0.5694	0.26378
H18	0.37848	0.53614	0.27277
H19	0.4808	0.647	0.74741
H20	0.43845	0.68015	0.75146
H21	0.41157	0.51089	0.6386
H22	0.47267	0.42382	0.57064
H23	0.58332	0.59569	0.18083
H24	0.63352	0.6031	0.2931
H25	0.60577	0.61504	0.63593

Supplementary Table 6 | Atomistic coordinates for the AB-stacking mode of TPB-DMTP-COF optimized by using DFTB+ method (space group $P\bar{3}$, $a = b = 36.6669 \text{ \AA}$, $c = 6.3984 \text{ \AA}$, $\alpha = \beta = 90^\circ$ and $\gamma = 120^\circ$)

Atom	x/a	y/b	z/c
C1	-0.02536	1.01757	0.28511
C2	-0.04381	0.97412	0.28508
H3	-0.04557	1.03158	0.28509
C4	0.35879	0.6492	0.28354
C5	0.37713	0.69264	0.28352
H6	0.37907	0.63528	0.28346
C7	0.05285	0.96336	0.28479
C8	0.0942	0.98559	0.21462
C9	0.11954	0.96807	0.21466
C10	0.10499	0.92699	0.28403
C11	0.06336	0.90477	0.35495
C12	0.03825	0.92247	0.35504
H13	0.10686	1.01777	0.15817
H14	0.15175	0.98656	0.15776
H15	0.05064	0.87262	0.41222
H16	0.00597	0.90402	0.41211
C17	0.21689	0.79424	0.29372
N18	0.20125	0.75422	0.27727
C19	0.11587	0.87188	0.27054
N20	0.13164	0.91194	0.28544
H21	0.25122	0.81572	0.3123
H22	0.08151	0.85044	0.25343
C23	0.2803	0.70309	0.28308
C24	0.23917	0.68102	0.20965
C25	0.21384	0.69854	0.20837
C26	0.22804	0.73933	0.28129
C27	0.26945	0.7614	0.35537
C28	0.29462	0.74377	0.35583
H29	0.22676	0.64902	0.15086
H30	0.1818	0.68017	0.14896
H31	0.28198	0.79343	0.414
H32	0.32676	0.76213	0.41498
C33	0.19135	0.81387	0.28843
C34	0.14791	0.79004	0.26255
C35	0.12325	0.80852	0.25642
C36	0.14137	0.85218	0.27523
C37	0.1848	0.87596	0.30065
C38	0.20947	0.85753	0.30781

H39	0.13364	0.75607	0.24684
O40	0.07894	0.78522	0.23067
H41	0.19956	0.9101	0.31707
O42	0.25368	0.88062	0.33811
C43	0.06174	0.74011	0.22355
C44	0.27176	0.92592	0.34297
H45	0.07022	0.7287	0.36808
H46	0.0721	0.73056	0.07864
H47	0.02743	0.72616	0.21793
H48	0.2908	0.93945	0.19749
H49	0.29222	0.93804	0.48436
H50	0.24723	0.93489	0.35146

Supplementary Table 7 | Atomistic coordinates for the refined unit cell parameters for TPB-DMTP-COF via Pawley refinement (space group $P6$, $a = b = 37.1541 \text{ \AA}$, $c = 3.5378 \text{ \AA}$, $\alpha = \beta = 90^\circ$ and $\gamma = 120^\circ$)

Atom	x/a	y/b	z/c
C1	0.28981	0.64218	0.50948
C2	0.31459	0.62381	0.50941
C3	0.24366	0.61623	0.51129
C4	0.36807	0.58669	0.38149
C5	0.39171	0.56757	0.38659
C6	0.43264	0.5889	0.52061
C7	0.44902	0.63002	0.64034
C8	0.42488	0.64857	0.64092
N9	0.45939	0.5724	0.51985
C10	0.44488	0.53315	0.57564
C11	0.47293	0.51623	0.56578
C12	0.45731	0.47302	0.55581
C13	0.48409	0.45714	0.56395
O14	0.41445	0.44831	0.5392
C15	0.6025	0.59337	0.4059
H16	0.29983	0.59004	0.50963
H17	0.33674	0.5694	0.26378
H18	0.37848	0.53614	0.27277
H19	0.4808	0.647	0.74741
H20	0.43845	0.68015	0.75146
H21	0.41157	0.51089	0.6386
H22	0.47267	0.42382	0.57064
H23	0.58332	0.59569	0.18083
H24	0.63352	0.6031	0.2931
H25	0.60577	0.61504	0.63593

Supplementary Table 8 | Atomistic coordinates for the refined unit cell parameters for TPB-DMTP-COF via Rietveld refinement (space group $P6$, $a = b = 36.4594 \text{ \AA}$; $c = 3.5239 \text{ \AA}$, $\alpha = \beta = 90^\circ$ and $\gamma = 120^\circ$)

Atom	x/a	y/b	z/c
C1	0.28981	0.64218	0.50948
C2	0.31459	0.62381	0.50941
C3	0.24366	0.61623	0.51129
C4	0.36807	0.58669	0.38149
C5	0.39171	0.56757	0.38659
C6	0.43264	0.5889	0.52061
C7	0.44902	0.63002	0.64034
C8	0.42488	0.64857	0.64092
N9	0.45939	0.5724	0.51985
C10	0.44488	0.53315	0.57564
C11	0.47293	0.51623	0.56578
C12	0.45731	0.47302	0.55581
C13	0.48409	0.45714	0.56395
O14	0.41445	0.44831	0.5392
C15	0.6025	0.59337	0.4059
H16	0.29983	0.59004	0.50963
H17	0.33674	0.5694	0.26378
H18	0.37848	0.53614	0.27277
H19	0.4808	0.647	0.74741
H20	0.43845	0.68015	0.75146
H21	0.41157	0.51089	0.6386
H22	0.47267	0.42382	0.57064
H23	0.58332	0.59569	0.18083
H24	0.63352	0.6031	0.2931
H25	0.60577	0.61504	0.63593

Supplementary Table 9 | Observed ethynyl content (X value) of $[\text{HC}\equiv\text{C}]_x\text{-TPB-DMTP-COFs}$ based on the standard curves using relative areas of $\text{C}\equiv\text{C}$ (2120 cm^{-1}) and CC-H (3300 cm^{-1}) bands to the common bands of C=N of imine linkages and C-C of aromatic rings

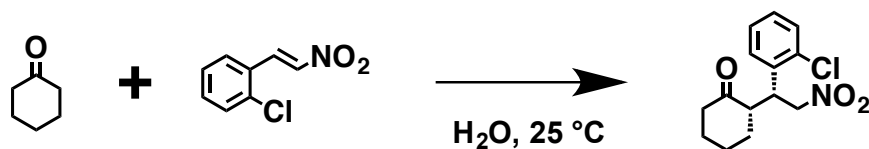
COFs	Relative Area (2120 cm^{-1})	Observed X value	Relative Area (3300 cm^{-1})	Observed X value
$[\text{HC}\equiv\text{C}]_{0.17}\text{-TPB-DMTP-COF}$	0.68	17.0	4.94	17.3
$[\text{HC}\equiv\text{C}]_{0.34}\text{-TPB-DMTP-COF}$	1.37	34.2	9.53	33.4
$[\text{HC}\equiv\text{C}]_{0.50}\text{-TPB-DMTP-COF}$	1.99	49.6	14.6	51.2
$[\text{HC}\equiv\text{C}]_{1.0}\text{-TPB-DMTP-COF}$	4.01	100	28.5	100

Supplementary Table 10 | Elemental analysis of COFs

COFs		C (%)	H (%)	N (%)
TPB-DMTP-COF	Calcd.	79.57	5.14	7.14
	Found	78.34	5.77	6.35
[HC≡C] _{0.17} -TPB-DMTP-COF	Calcd.	79.98	5.03	7.00
	Found	78.32	5.57	6.25
[HC≡C] _{0.34} -TPB-DMTP-COF	Calcd.	80.37	4.93	6.86
	Found	78.31	5.64	5.94
[HC≡C] _{0.50} -TPB-DMTP-COF	Calcd.	80.75	4.84	6.73
	Found	77.89	5.28	5.81
[S-Py] _{0.17} -TPB-DMTP-COF	Calcd.	76.90	5.31	10.55
	Found	74.11	5.28	10.05
[S-Py] _{0.34} -TPB-DMTP-COF	Calcd.	74.78	5.46	13.27
	Found	72.22	5.43	12.38
[S-Py] _{0.50} -TPB-DMTP-COF	Calcd.	73.04	5.57	15.49
	Found	70.20	5.45	14.75

Supplementary Table 11 | The Cu content of [S-Py]_x-TPB-DMTP-COFs determined using ICP-AES

COFs	Cu content (wt%)
[S-Py] _{0.17} -TPB-DMTP-COFs	0.009
[S-Py] _{0.34} -TPB-DMTP-COFs	0.004
[S-Py] _{0.50} -TPB-DMTP-COFs	0.004

Supplementary Table 12 | Michael reactions in the presence of Cu, CuI and CuSO₄•5H₂O

COFs	Cu species (5 mg)	Time (h)	Yield (%)	ee (%)	dr
-	Cu powder	72	0	-	-
-	CuI	72	0	-	-
-	CuSO ₄ •5H ₂ O	72	0	-	-
TPB-DMTP-COF (13 mg = 10 mol%)	Cu powder	72	0	-	-
TPB-DMTP-COF (13 mg)	CuI	72	0	-	-
TPB-DMTP-COF (13 mg)	CuSO ₄ •5H ₂ O	72	0	-	-
[S-Py] _{0.17} -TPB-DMTP-COF (13 mg)	Cu powder	6	94	94	97/3
[S-Py] _{0.17} -TPB-DMTP-COF (13 mg)	CuI	6	92	94	97/3
[S-Py] _{0.17} -TPB-DMTP-COF (13 mg)	CuSO ₄ •5H ₂ O	6	92	94	97/3

Supplementary Table 13 | Heterogeneous organocatalysts based on porous materials for Michael or Aldol reactions reported to date

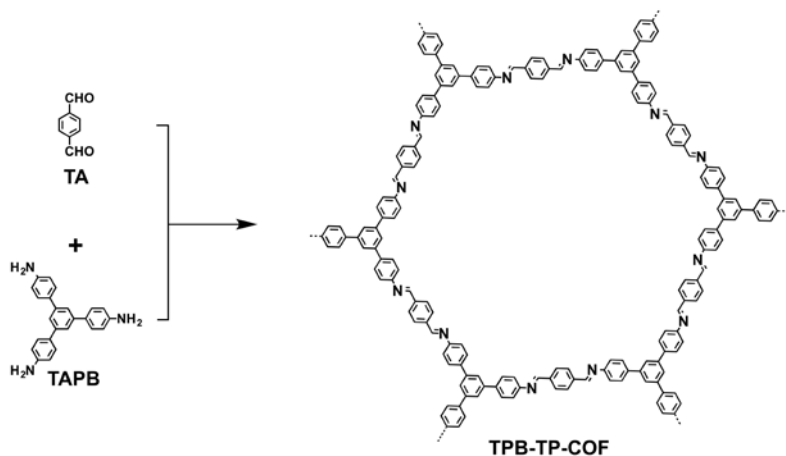
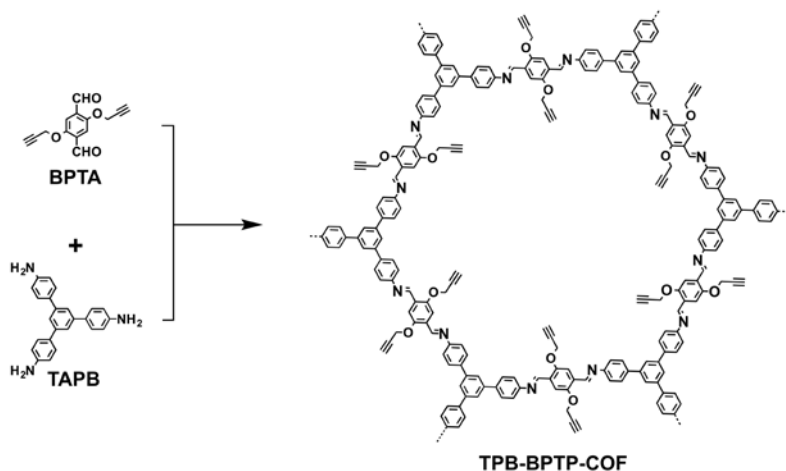
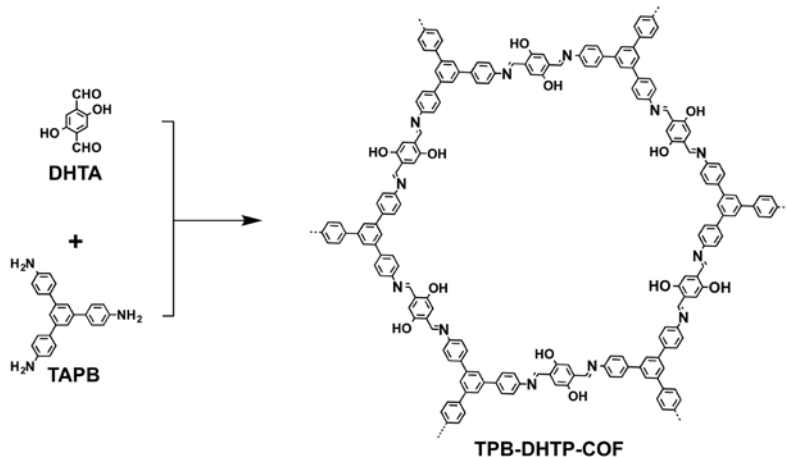
	Preparation Strategy	Catalytic Site	Reactions	Solvents	Catalyst (mol%)	Temp (°C)	Time (h)	Yield (%)	ee (dr)	Cycle Performance	Surface Area and Pore Size	Reference
Microporous COFs	pyrrolidine units covalently linked to pore walls			EtOH/H ₂ O	10	25	0.75	93	56% (60/40)	After 4 cycles, the ee and dr could be well maintained, the yield is slightly decreased from 93% to 86% and the reaction time is extended from 1 h to 4.6 h.	960 m ² g ⁻¹ 0.56 cm ³ g ⁻¹ Pore size 1.9 nm	<i>Chem. Commun.</i> 50 , 1292-1294, (2014).
				EtOH/H ₂ O	10	25	1	93	49% (70/30)			
				H ₂ O	10	25	36	trace	-			
Mesoporous COFs	pyrrolidine units covalently linked to pore walls			H ₂ O	10	25	12	95	92% (90/10)	After 5 cycles, the ee and dr are almost unchanged, the yield is slightly decreased from 95% to 92%, and the reaction time is extended from 6 h to 13 h. Meanwhile, the COF skeleton is perfectly retained (both crystallinity and porosity).	1979 m ² g ⁻¹ 1.13 cm ³ g ⁻¹ Pore size 3.1 nm	This work
				H ₂ O	10	25	6	95	94% (97/3)			
				H ₂ O	10	25	26	95	96% (94/6)			
Porous Organic Polymers	Jørgensen-Hayashi catalyst ¹⁻³ covalently linked to the amorphous porous polymer skeletons			EtOH/H ₂ O	10	25	1.5	99	98% (92/8)	After 4 cycles, the ee and dr can be well maintained, however, the yield is decreased from 96% to 39%, and the reaction time (to achieve 100% conversion) is extended from 2 h to 30 h.	881 m ² g ⁻¹ 0.50 cm ³ g ⁻¹ Micropore (49.7%) Mesopore (50.3%)	<i>Chem. Eur. J.</i> 18 , 6718-6723, (2012).
				EtOH/H ₂ O	10	25	2	96	99% (92/8)			
				EtOH/H ₂ O	10	25	6	86	97% (95/5)			

Porous Organic Polymers	proline copper complex linked to porous polymer networks			1	ClCH ₂ CH ₂ Cl	10 (5)*	60	18 (18)	90 (60)	40% (60%)	This catalyst could be reused 10 times. (No detailed data)	190 m ² g ⁻¹ 0.33 cm ³ g ⁻¹ pore size 10.9 nm	Chem. Eur. J. 20 , 5111-5120, (2014).
				2	ClCH ₂ CH ₂ Cl	10 (5)	60	18 (18)	92 (60)	40% (30%)			
				3	ClCH ₂ CH ₂ Cl	10 (5)	60	18 (2)	96 (72)	5% (3%)			
				4	ClCH ₂ CH ₂ Cl	10 (5)	60	18 (2)	96 (97)	85% (47%)			
Graphenes	proline adsorbed on Graphene oxide <i>via</i> hydrogen bonds			Acetone	30	30	6	96	79%	The cycled catalyst could maintain the yield and ee; however the activity issue is not mentioned.	Not mentioned	J. Catal. 298 , 138-147, (2013).	
				DMF	30	30	24	96	92% (93/7)				
Zeolites	proline adsorbed in zeolite			Acetone	30	25	5.5	78	-21%	The cycle performance is not mentioned.	209 m ² g ⁻¹	J. Catal. 243 , 442-445, (2006).	
				Acetone	30	25	40	18	-8%				
	proline covalently linked to the walls of MCM-41 ⁴			DMSO	20	90	24	55	70% (58/42)	After 3 cycles, the yield (from 55% to 50%) and dr (from 58/42 to 56/44) could be well maintained. (ee not mentioned)	MCM-41 (1030 m ² g ⁻¹) Surface area of the catalyst is not mentioned.	Adv. Synth. Catal. 347 , 1395-1403, (2005).	
				Toluene	20	90	24	60	70% (61/39)				
				DMSO	20	90	24	60	70% (95/5)				
				Toluene	20	90	24	55	70% (95/5)				
				DMSO	20	90	24	55	99% (95/5)				
				Toluene	20	90	24	55	99% (95/5)				
				MeCN	30%	25	96	20	63% (66/34)	After 1 recycle, the conversion is changed from 96% to 93% at same period of time, ee decreased from 67% to	MCM-41 (1030 m ² g ⁻¹) Surface area of the catalyst is not	J. Org. Chem. 72 , 9353-9356, (2007).	
				DMF	30%	25	61	80	82% (83/17)				

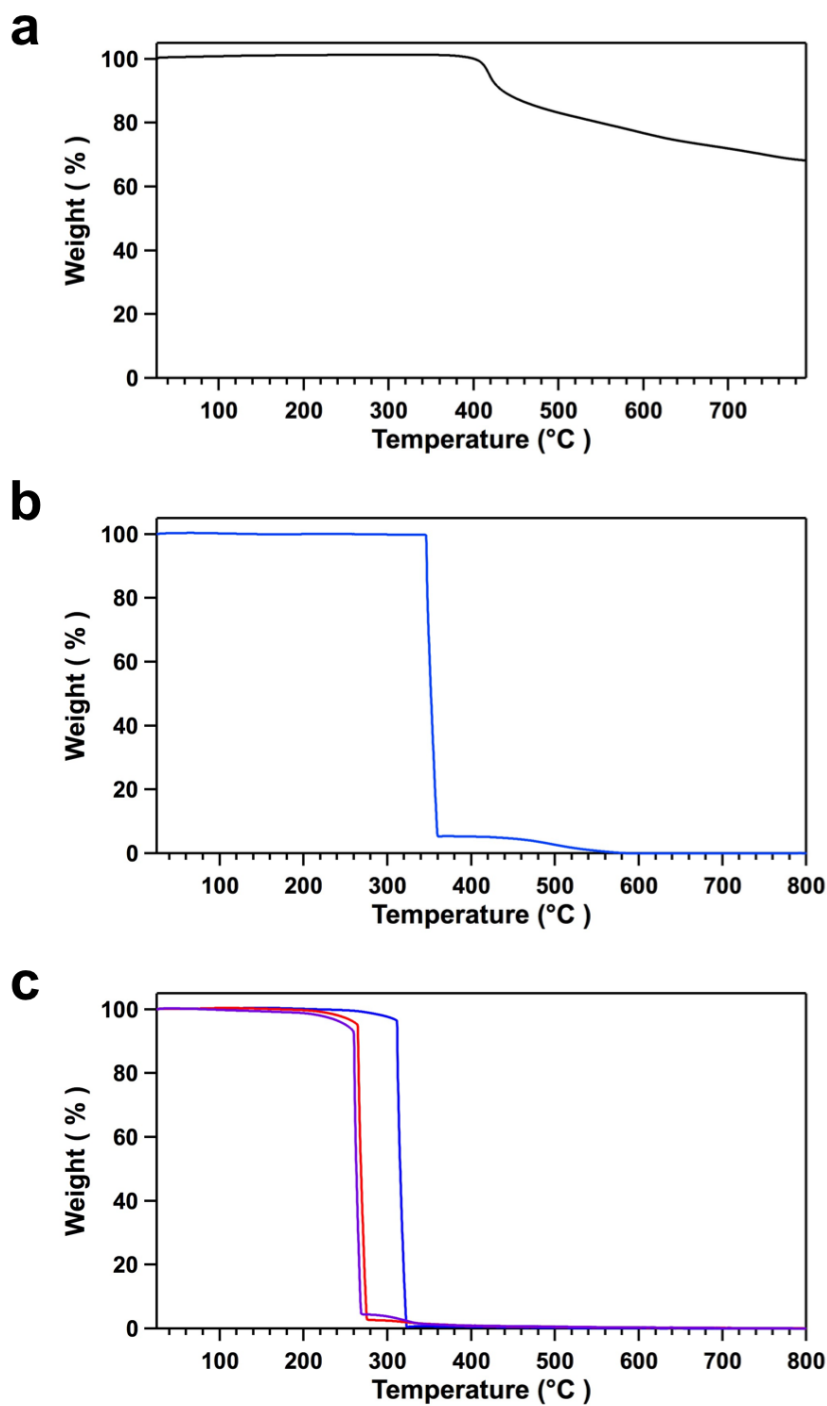
MOFs				Formamide	30	25	13	96	67% (66/34)	65%.	mentioned.	
				EtOH/H ₂ O	10	25	6	86	97% (95/5)			
	proline coordinated to a MOF (MIL-101) ⁵			Neat condition	10	25	24	66	69%	After 3 cycles, the reaction time was extended from 2 h to 48 h (14% starting material residue); yield was decreased from 66% to 48% and ee was decreased from 69% to 58%. In this last cycle, 11% of ligand was observed in the solution.	1420 m ² g ⁻¹ 0.73 cm ³ g ⁻¹ (before introducing catalyst: 3850 m ² g ⁻¹ 2.06 cm ³ g ⁻¹)	<i>J. Am. Chem. Soc.</i> 131 , 7524-7525, (2009).
				Neat condition	10	25	24	81	66% (80/20)			
				DMF	10	25	36	86	68% (83/17)			
				Neat condition	10	25	36	49	81%			
	proline coordinated to a Cd-TBT MOF			MeOH/H ₂ O	5	25	10 days	42	60%	After 3 cycles, the ee could be well maintained, and the yield is slightly decreased from 97% to 89% (10 days).	Not mentioned.	<i>J. Am. Chem. Soc.</i> 132 , 14321-14323, (2010).
				MeOH/H ₂ O	5	25	10 days	77	61%			
				MeOH/H ₂ O	5	25	10 days	97	58%			
	proline coordinated to an IRMOF-pro			Acetone	100	25	40	-	29%	After 3 cycles, the reaction time was extended from 30 h to 72 h.	138 m ² g ⁻¹	<i>J. Am. Chem. Soc.</i> 133 , 5806-5809, (2011).
				Cyclopentanone	100	25	30	-	14% (75/25)			

* Data in the parenthesis refer to the result of the small molecule catalyst.

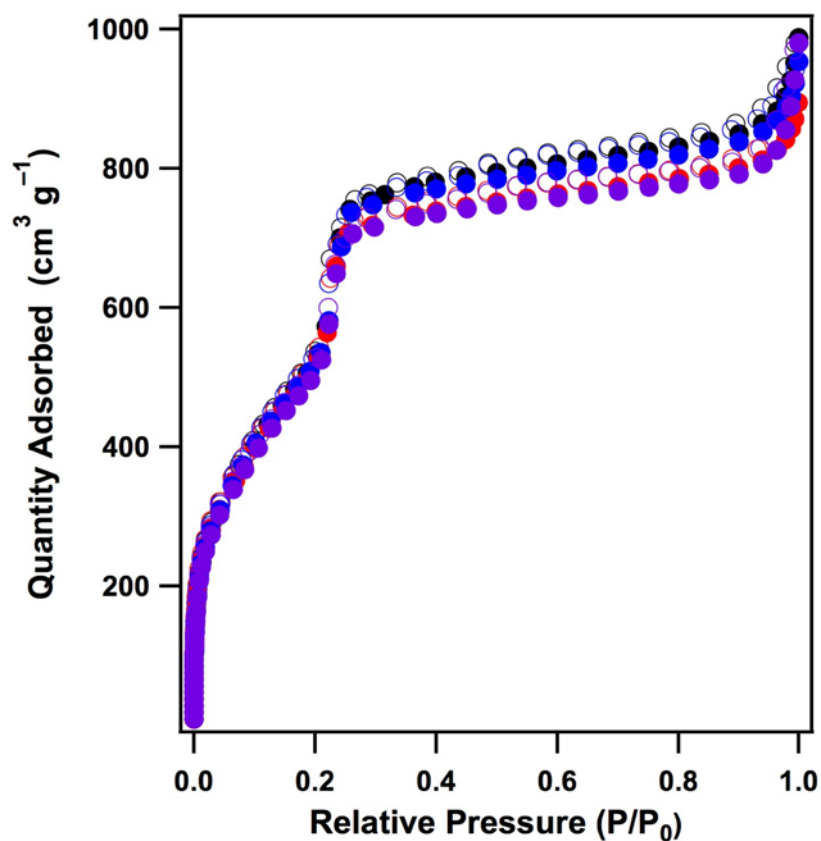
Supplementary Figures

a**b****c**

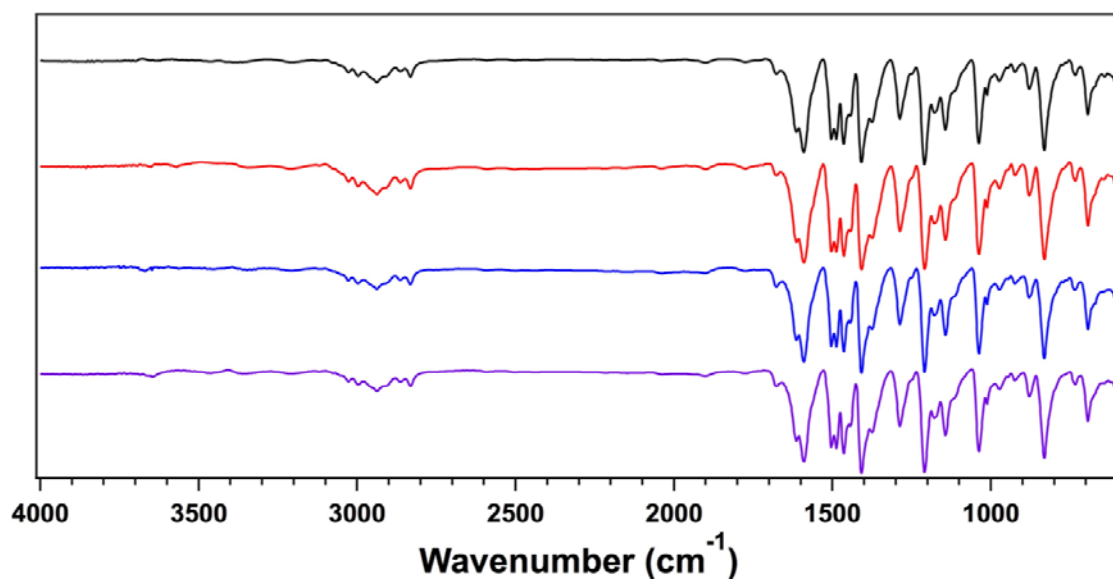
Supplementary Figure 1 | Schematics of (a) TPB-TP-COF without methoxy groups on the phenyl edges, (b) TPB-BPTP-COF ($\equiv [\text{HC}\equiv\text{C}]_{1.0}$ -TPB-DMTP-COF) with 100% ethynyl groups on the phenyl edges and (c) TPB-DHTP-COF with phenol groups on the edges.



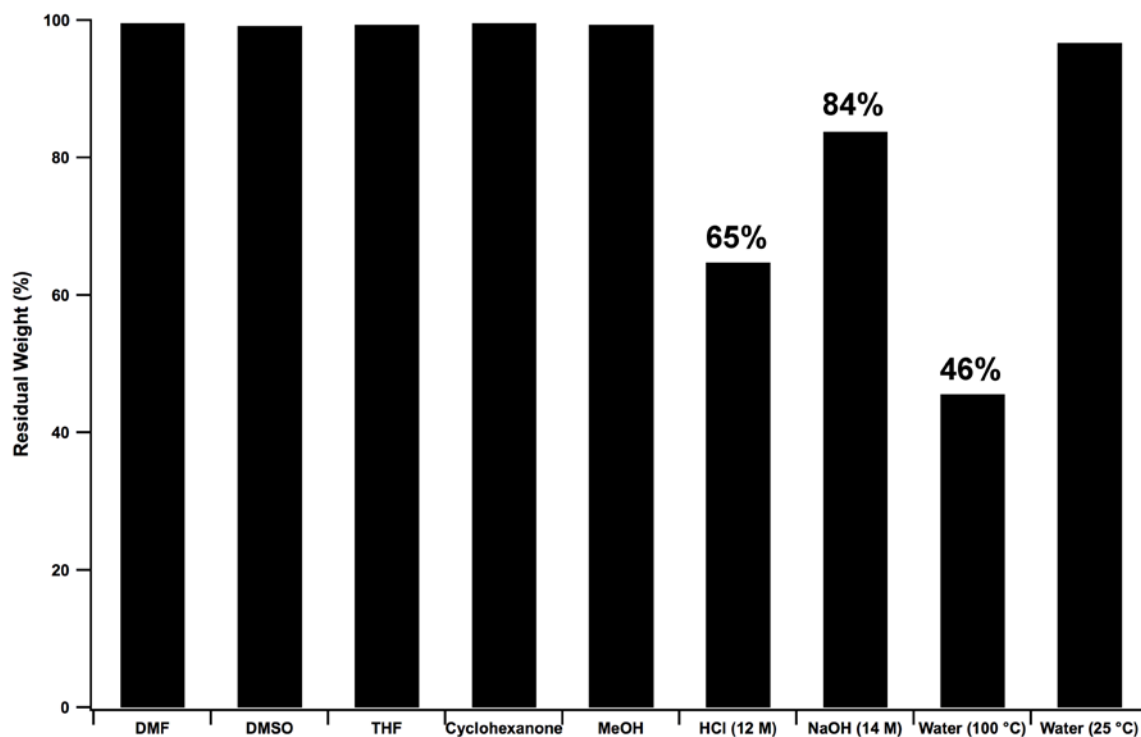
Supplementary Figure 2 | TGA curves of (a) TPB-DMTP-COF under N₂, (b) TPB-DMTP-COF under O₂ and (c) [S-Py]_x-TPB-DMTP-COFs under O₂ (blue: X = 0.17, red: X = 0.34, purple: X = 0.50).



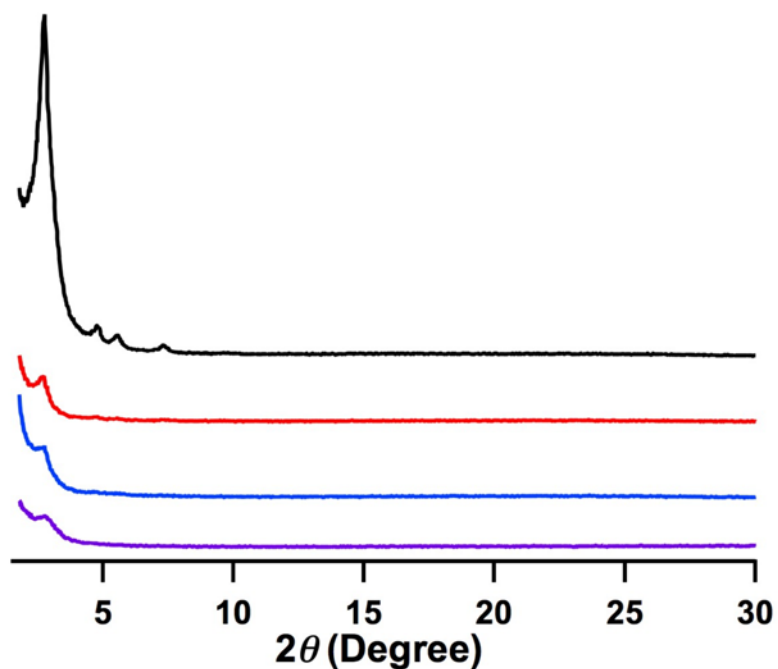
Supplementary Figure 3 | Nitrogen sorption isotherm profiles of TPB-DMTP-COF (black) upon treatment in 12 M HCl (red), boiling water (blue) and 14 M NaOH (purple) for one week. The filled circles represent adsorption and the open circles represent desorption.



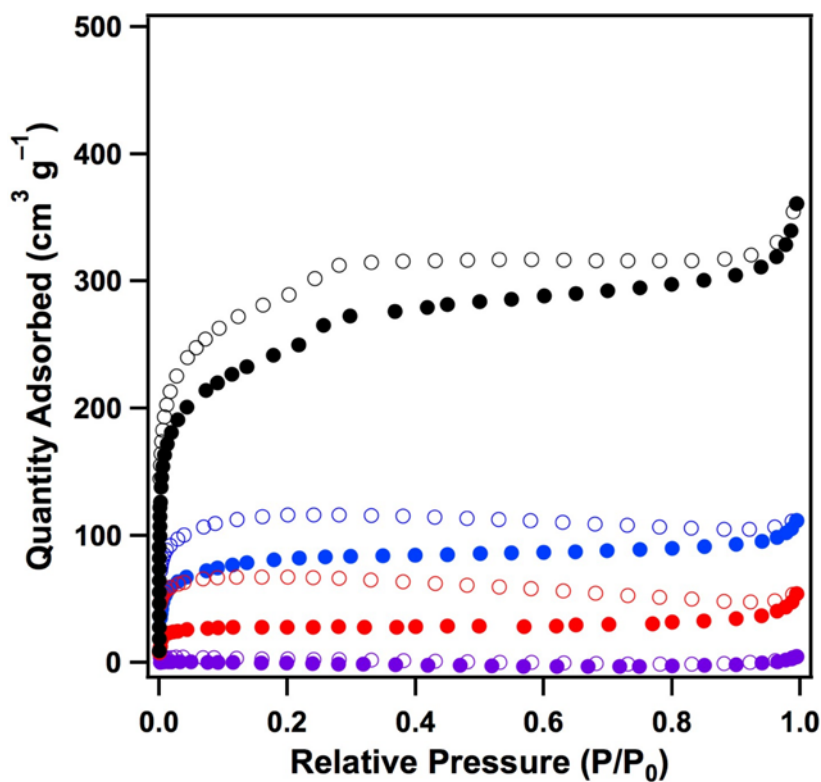
Supplementary Figure 4 | FT-IR spectra of TPB-DMTP-COF (black) upon treatment in 12 M HCl (red), boiling water (blue) and 14 M NaOH (purple) for one week.



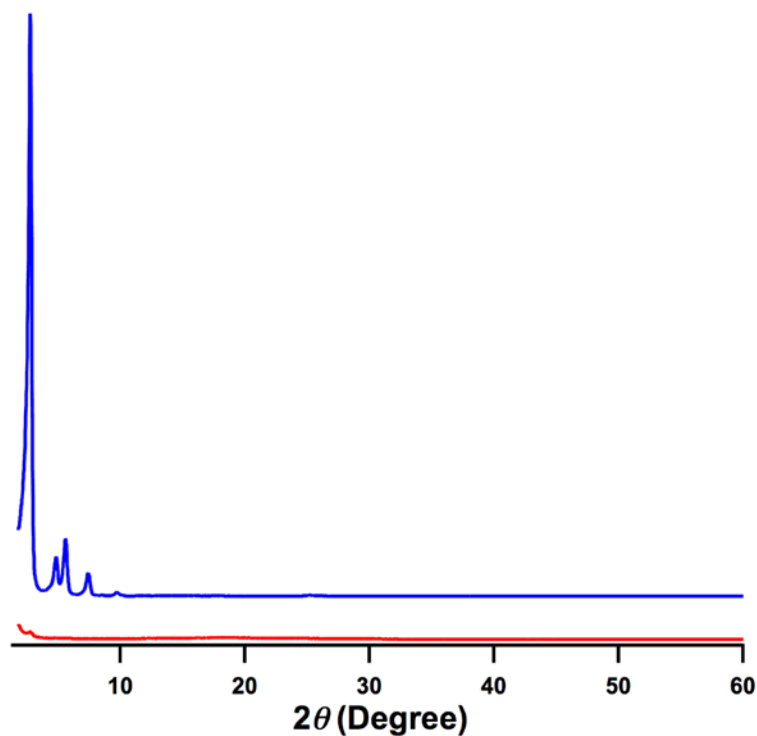
Supplementary Figure 5 | Weight loss of TPB-TP-COF in different solvents for one week.



Supplementary Figure 6 | PXRD patterns of TPB-DHTP-COF after treatment in different solvents for one week (black: as-synthesized, red: 12 M HCl, blue: boiling water, purple: 14 M NaOH).

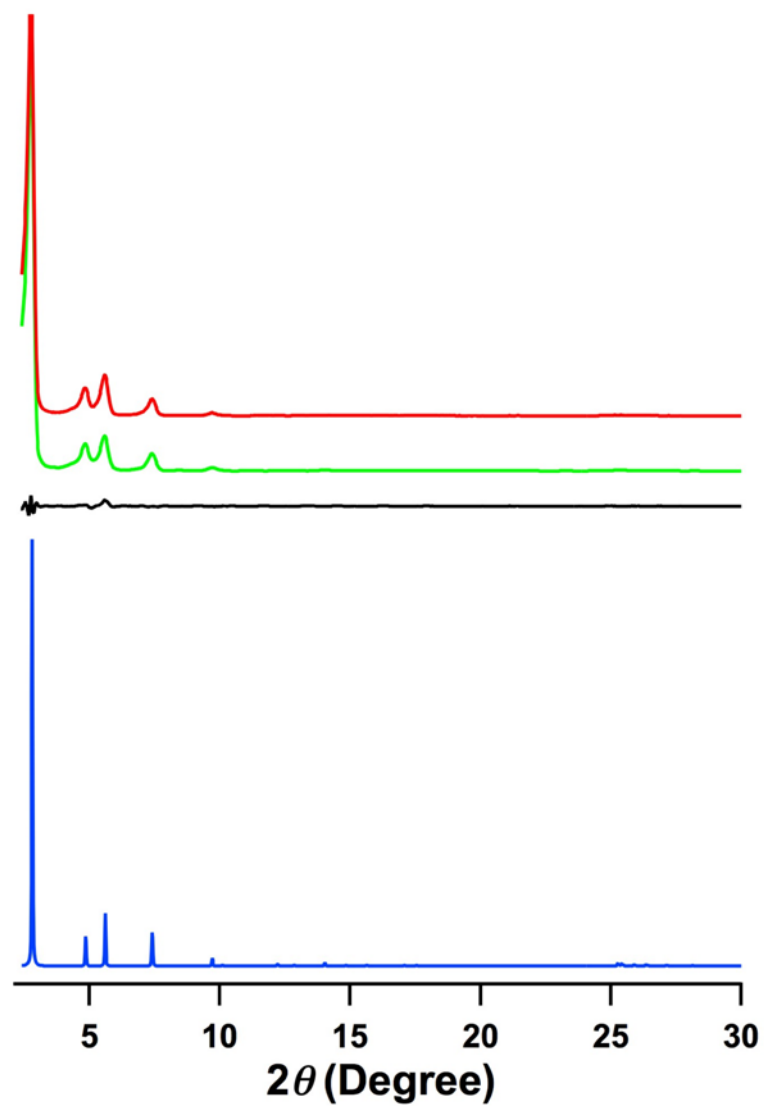


Supplementary Figure 7 | Nitrogen sorption isotherm profiles of TPB-DHTP-COF after treatment in different solvents for one week (black: as-synthesized, red: 12 M HCl, blue: boiling water, purple: 14 M NaOH).

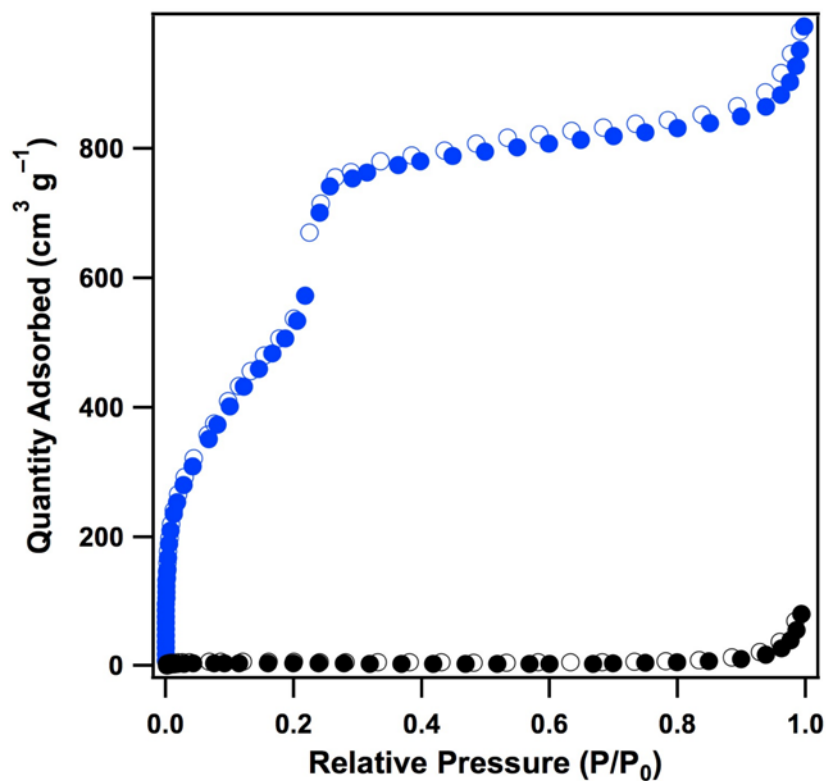


Supplementary Figure 8 | PXRD patterns of TPB-DMTP-COF (blue) and TPB-TP-COF (red).

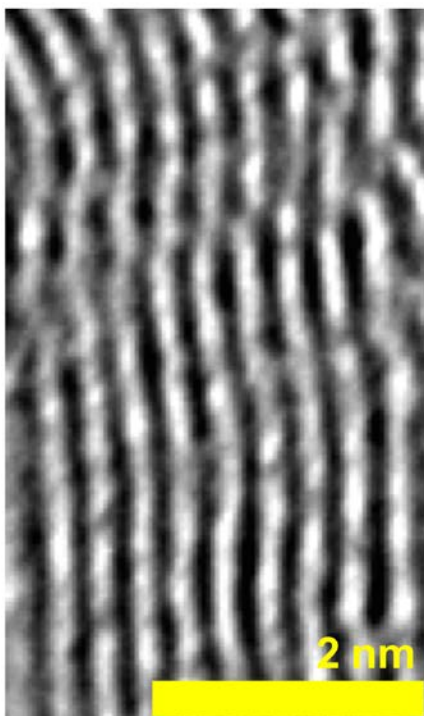
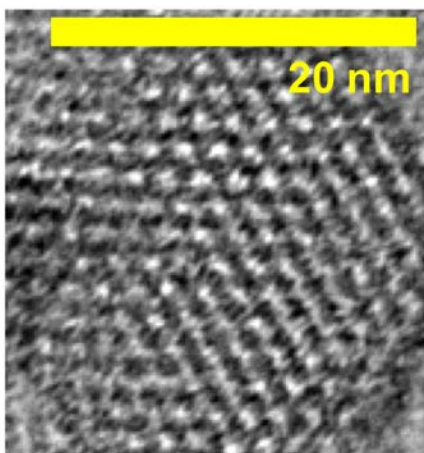
The PXRD intensity of the main peak at 2.76° is 135×10^3 cps for TPB-DMTP-COF. The intensity of the peak at 2.76° is only 1.7×10^3 cps for TPB-TP-COF under otherwise identical PXRD conditions.



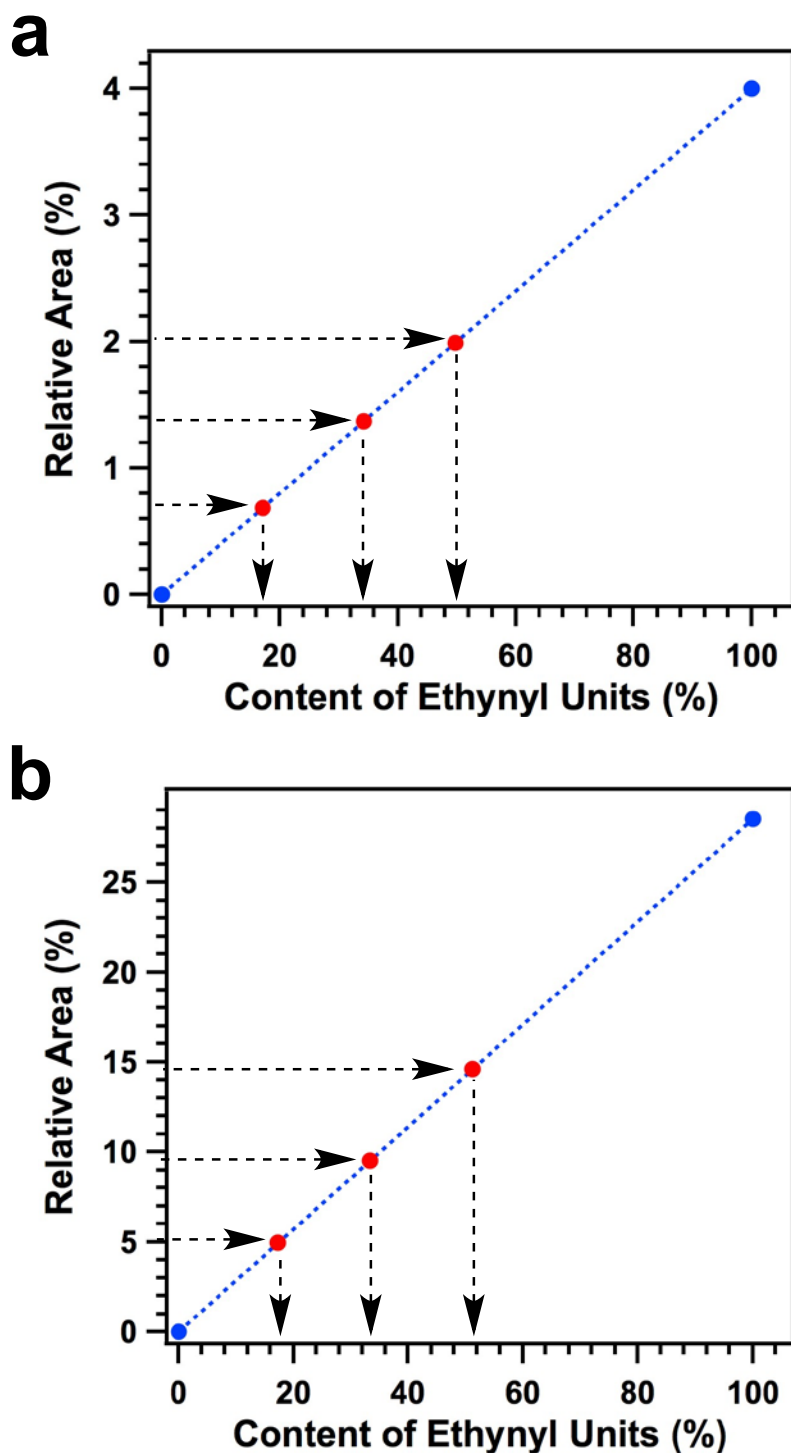
Supplementary Figure 9 | PXRD profiles of TPB-DMTP-COF (red: experimentally observed, green: Rietveld refined, black: their difference, blue: simulated using AA-stacking mode).



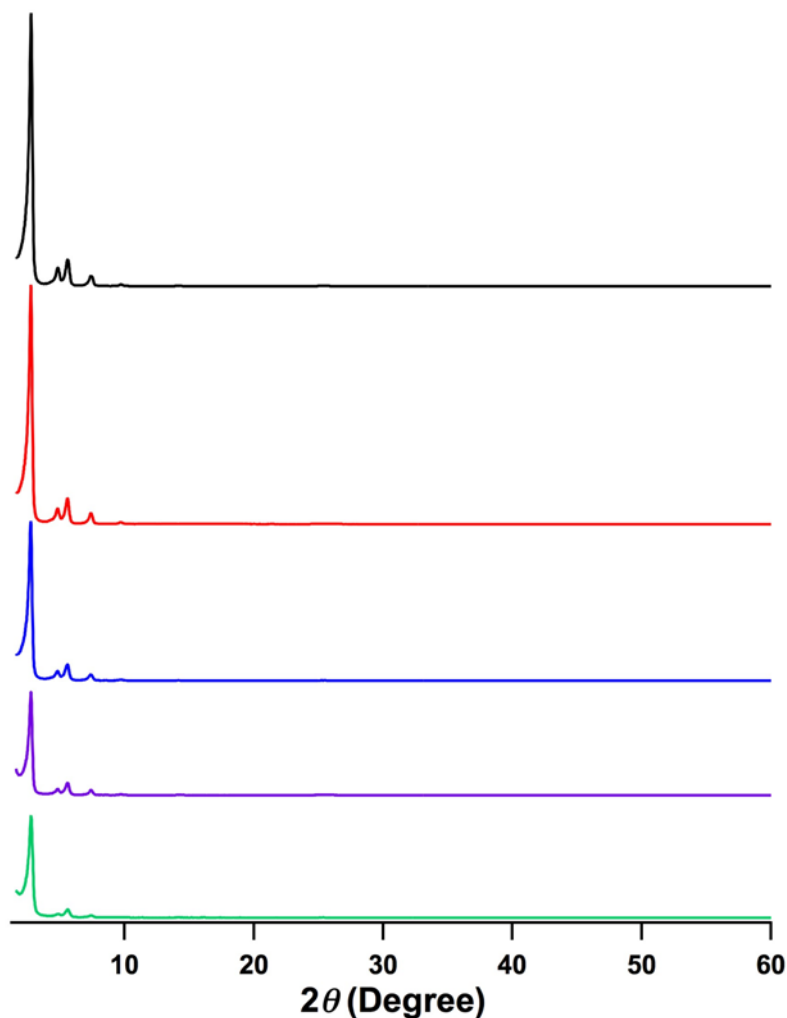
Supplementary Figure 10 | Nitrogen sorption isotherm profiles of TPB-DMTP-COF (blue) and TPB-TP-COF (black). The filled circles represent adsorption and the open circles represent desorption.



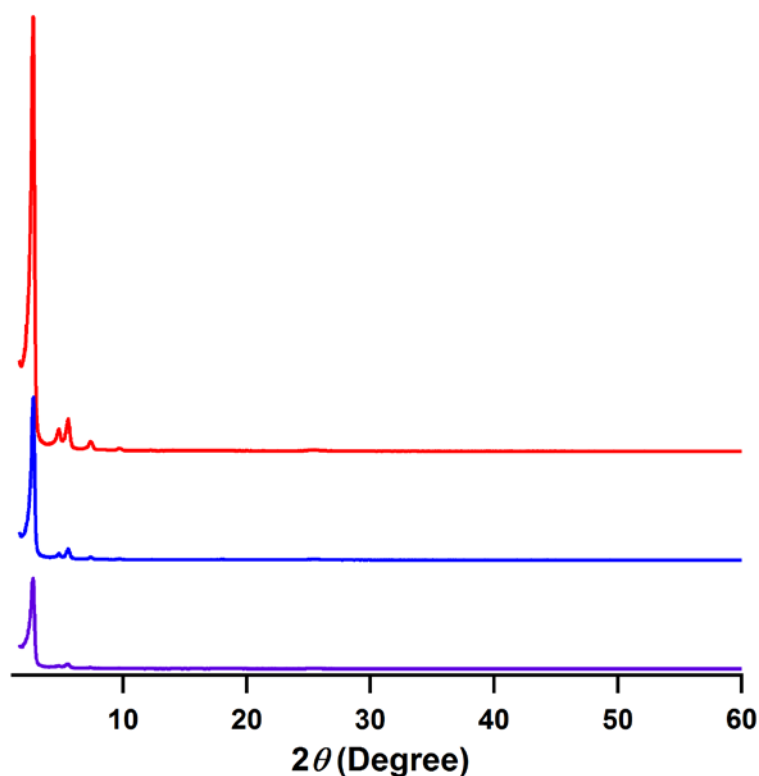
Supplementary Figure 11 | HR TEM images of TPB-DMTP-COF (upper: hexagonal texture, lower: stacking layers).



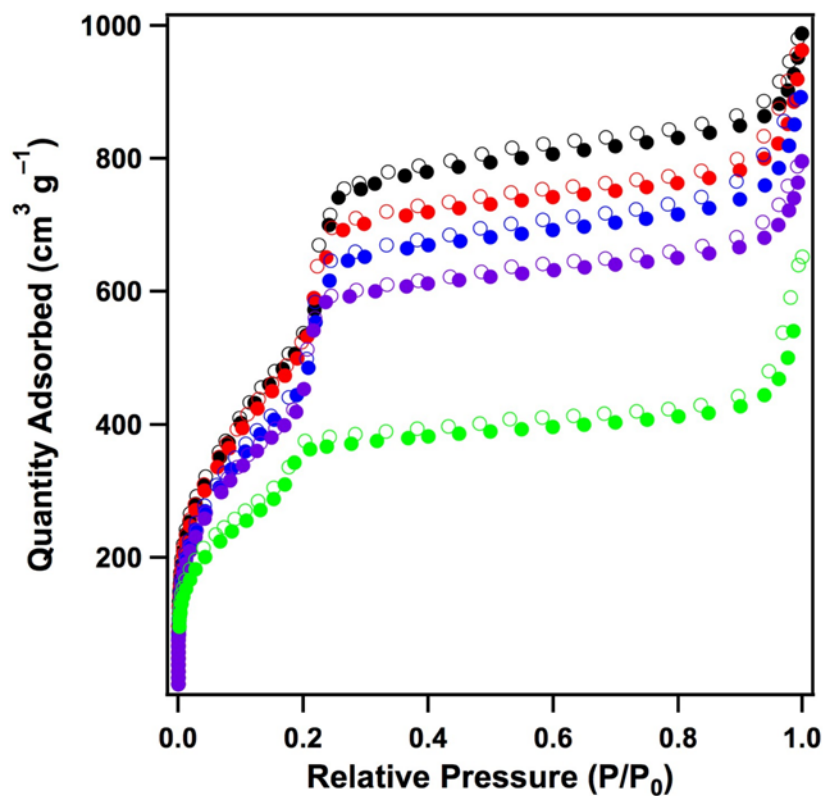
Supplementary Figure 12 | Standard curves based on the relative areas of (a) C≡C band (2120 cm^{-1}) and (b) the CC-H in ethynyl group (3300 cm^{-1}) to the common bands of C=N and C-C stretch in imine linkages and aromatic rings (from 1658 to 1533 cm^{-1}) versus the content of ethynyl groups for the estimation of the [HC≡C] content of [HC≡C]_x-TPB-DMTP-COFs.



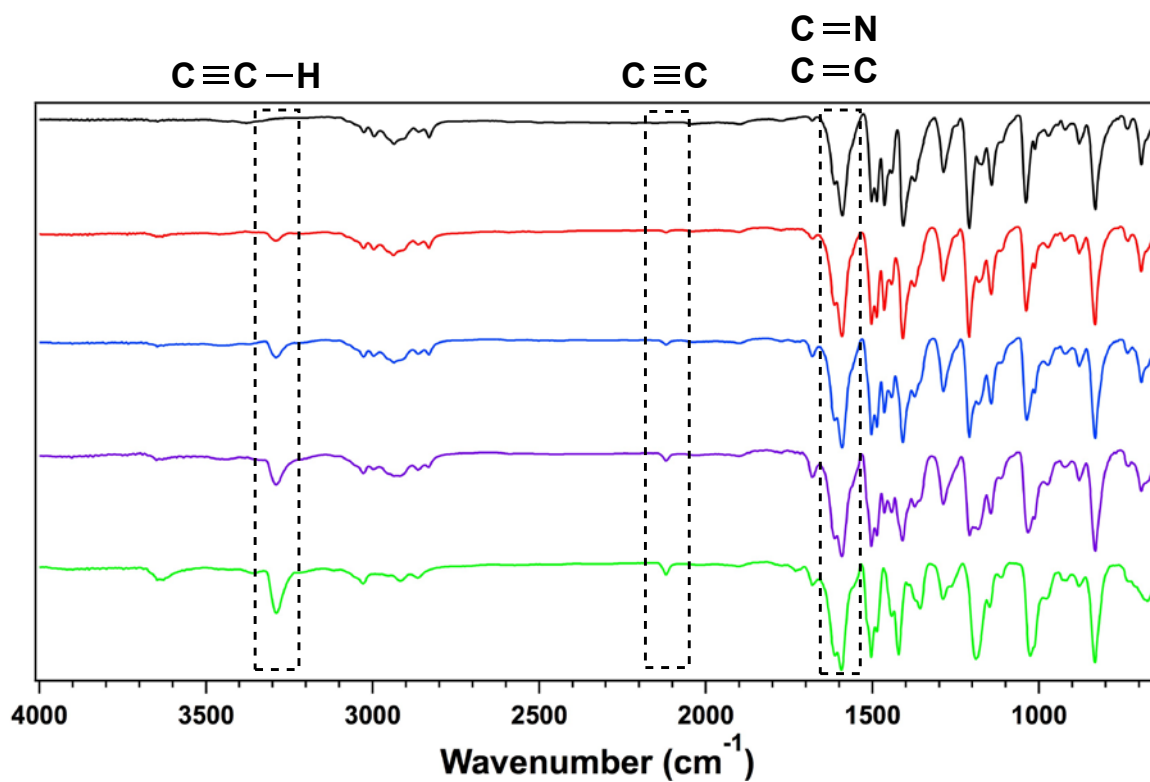
Supplementary Figure 13 | PXRD patterns of TPB-DMTP-COF (black curve) and $[\text{HC}\equiv\text{C}]_x$ -TPB-DMTP-COFs (red : $X = 0.17$, blue : $X = 0.34$, purple : $X = 0.50$, green : $X = 1.0$). The relatively low PXRD intensity observed for the $[\text{HC}\equiv\text{C}]_x$ -TPB-DMTP-COFs when X was increased from 0.17 to 0.34, 0.50 and 1.0 is attributed to the amorphous nature of the chains appended to the pore walls, a same phenomenon as previously reported for other wall surface-modified COFs⁶⁻⁸.



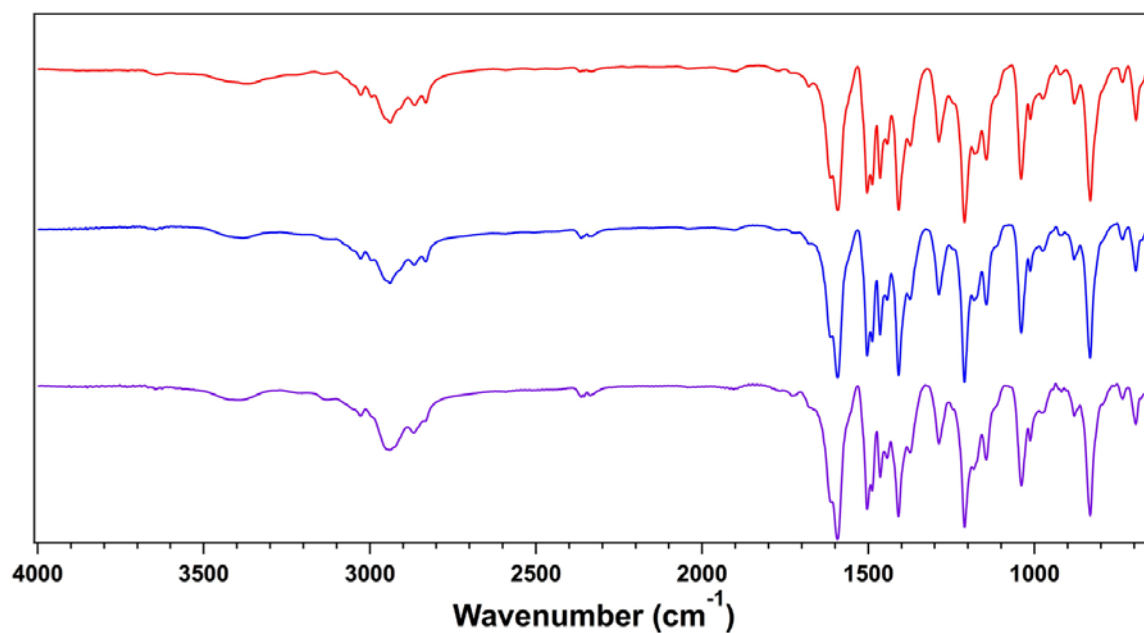
Supplementary Figure 14 | PXRD patterns of chiral $[S\text{-Py}]_x\text{-TPB-DMTP-COFs}$ (red: $X = 0.17$, blue: $X = 0.34$, purple: $X = 0.50$). The relatively low PXRD intensity observed for the $[Py]_x\text{-TPB-DMTP-COFs}$ when X was increased from 0.17 to 0.34 and 0.50 is attributed to the amorphous nature of the chains appended to the pore walls, a same phenomenon as previously reported for other wall surface-modified COFs⁶⁻⁸.



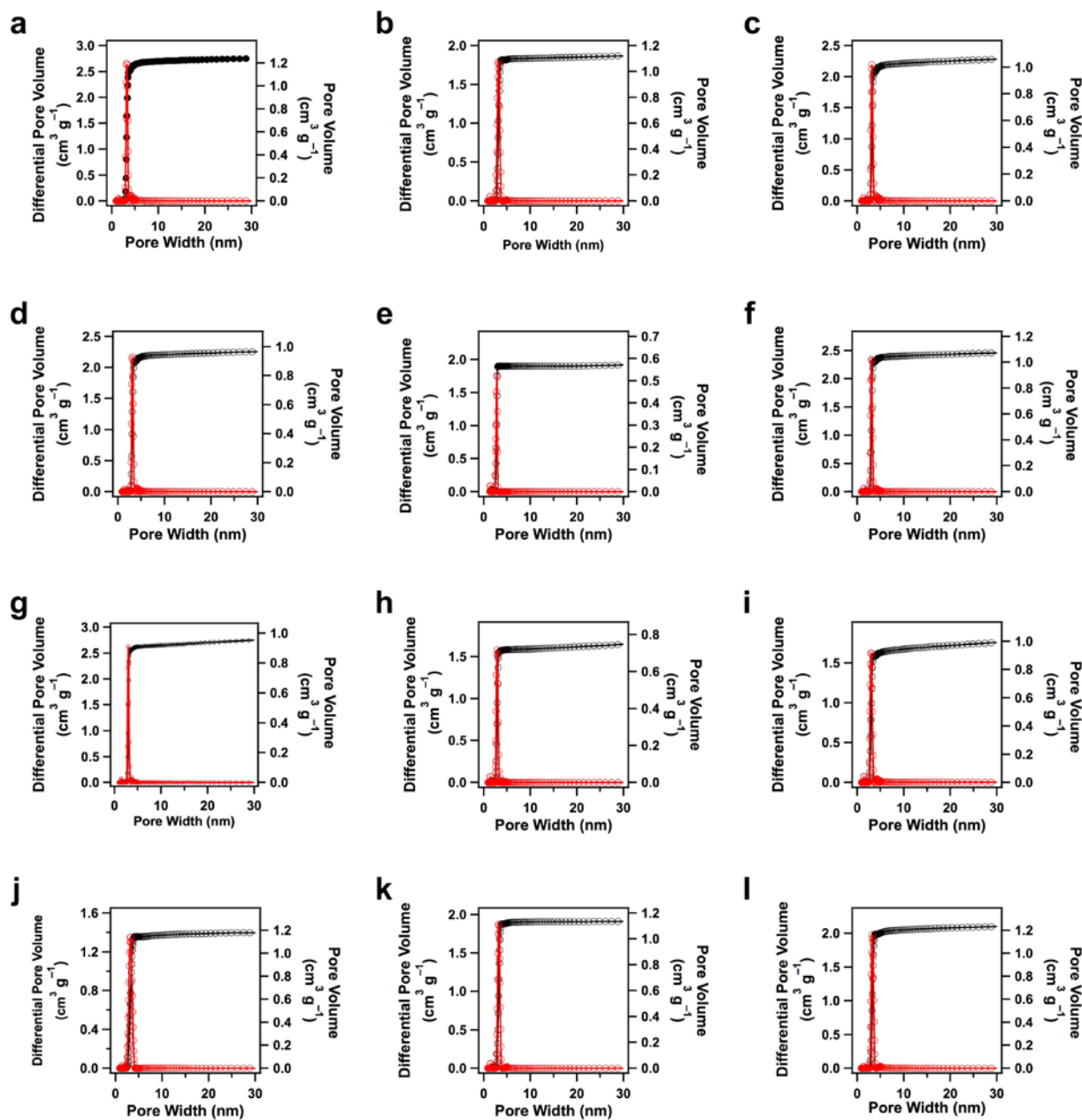
Supplementary Figure 15 | Nitrogen sorption isotherm profiles of TPB-DMTP-COF (black) and [HC≡C]_x-TPB-DMTP-COFs (red: X = 0.17, blue: X = 0.34, purple: X = 0.50, green: X = 1.0). The filled circles represent adsorption and the open circles represent desorption.



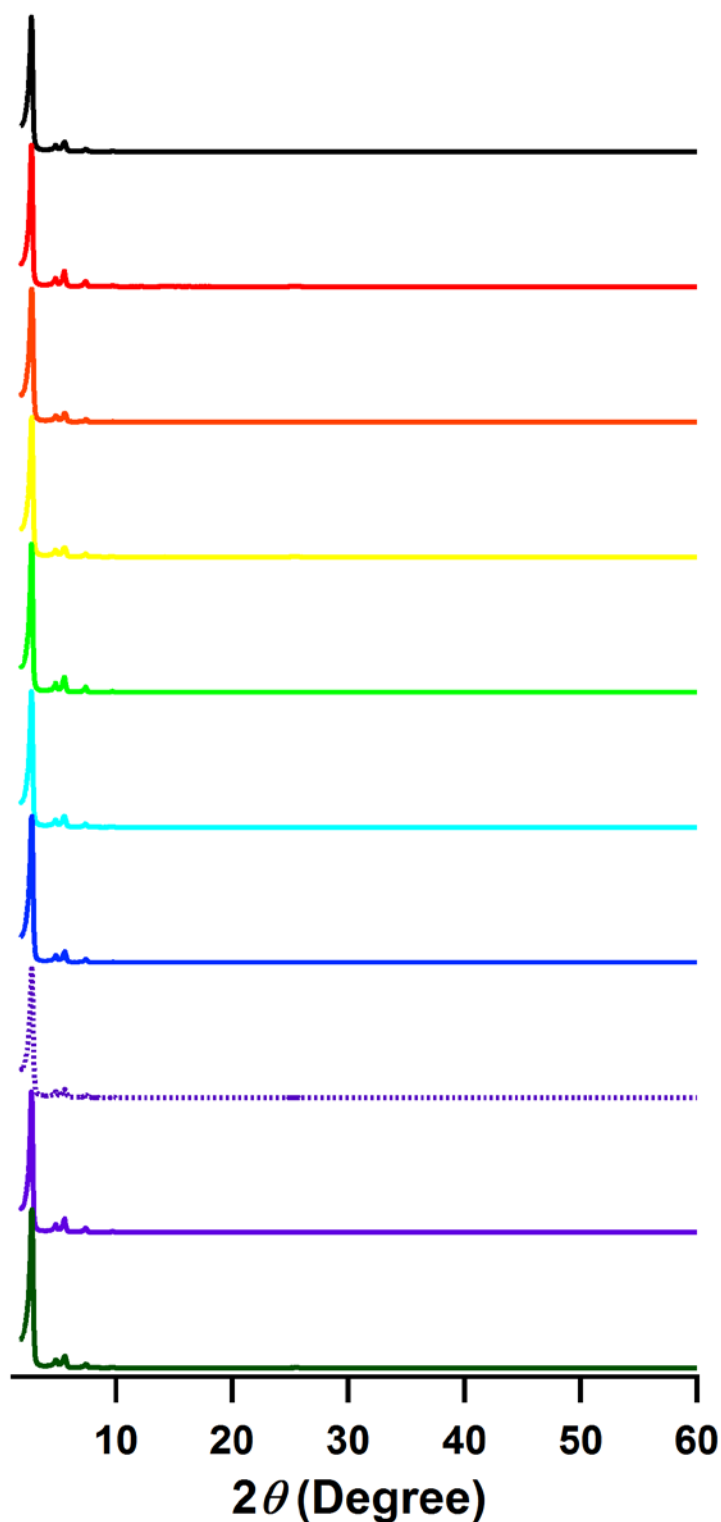
Supplementary Figure 16 | FT-IR spectra of TPB-DMTP-COF (black curve) and $[\text{HC}\equiv\text{C}]_x$ -TPB-DMTP-COFs (red: $x = 0.17$, blue: $x = 0.34$, purple: $x = 0.50$, green: $x = 1.0$).



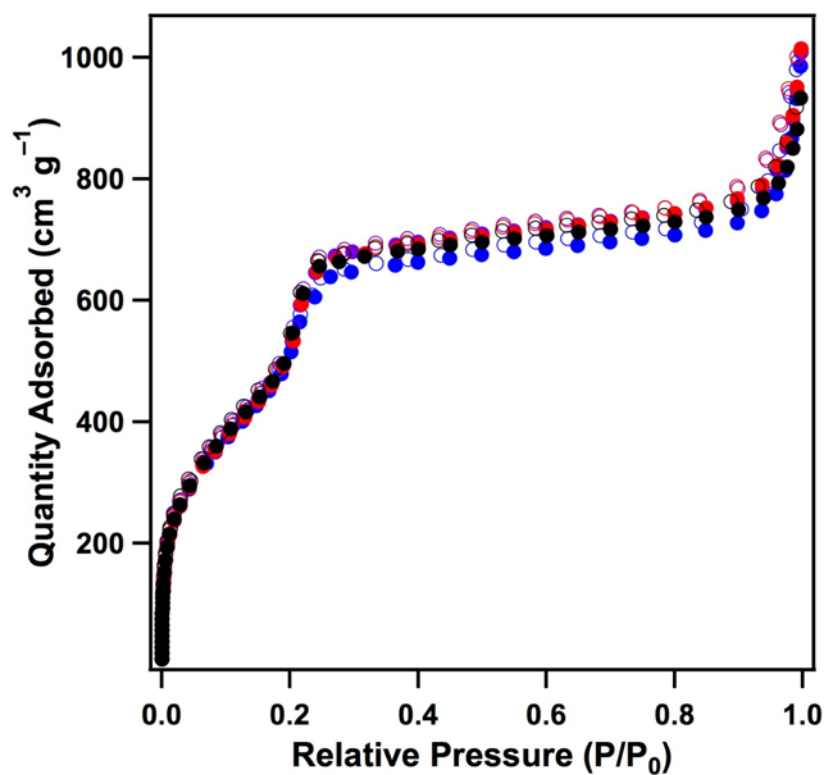
Supplementary Figure 17 | FT-IR spectra of $[S\text{-Py}]_x\text{-TPB-DMTP-COFs}$ (red: $x = 0.17$, blue: $x = 0.34$, purple: $x = 0.50$).



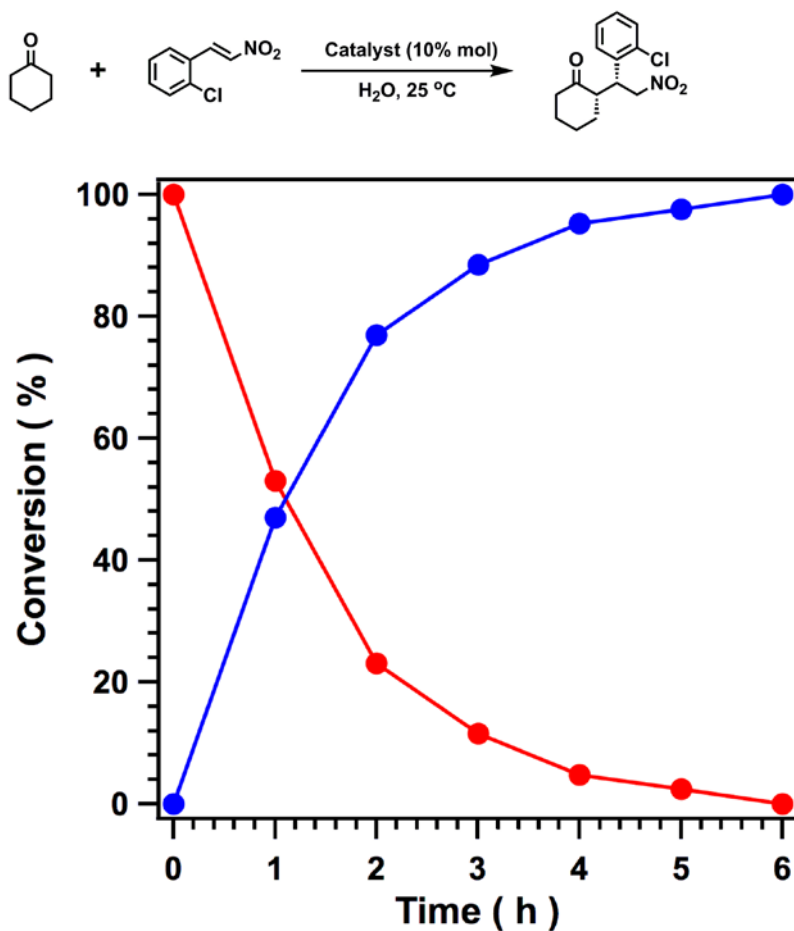
Supplementary Figure 18 | Pore size distribution (red dots) and cumulative pore volume (black dots) profiles of the COFs (**a**: TPB-DMTP-COF, **b**: [HC≡C]_{0.17}-TPB-DMTP-COF, **c**: [HC≡C]_{0.34}-TPB-DMTP-COF, **d**: [HC≡C]_{0.50}-TPB-DMTP-COF, **e**: [HC≡C]_{1.0}-TPB-DMTP-COF, **f**: [S-Py]_{0.17}-TPB-DMTP-COF, **g**: [S-Py]_{0.34}-TPB-DMTP-COF, **h**: [S-Py]_{0.50}-TPB-DMTP-COF, **i**: five-cycled [S-Py]_{0.17}-TPB-DMTP-COF, **j**: TPB-DMTP-COF treated with 12 M HCl for one week, **k**: TPB-DMTP-COF treated with boiling water for one week, **l**: TPB-DMTP-COF treated with 14 M NaOH for one week).



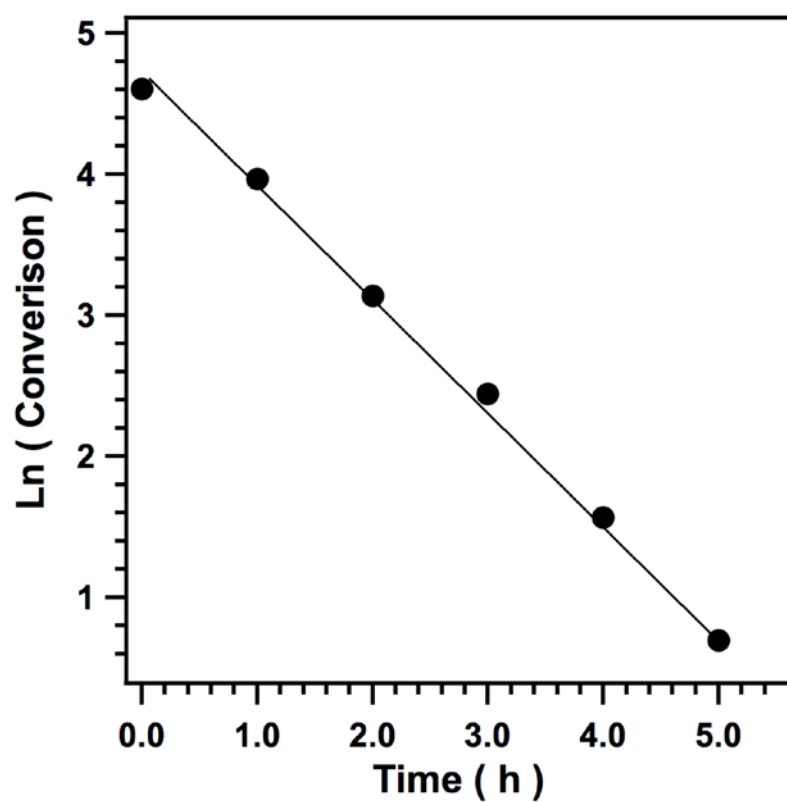
Supplementary Figure 19 | PXRD patterns of $[S-Py]_{0.17}$ -TPB-DMTP-COFs after treated in different solvents for one week (black: as-synthesised, red: DMF, orange: DMSO, yellow: THF, green: MeOH, sky blue: cyclohexanone, blue: 12 M HCl, dashed purple: boiling water (100 °C), purple: water, deep green: 14 M NaOH).



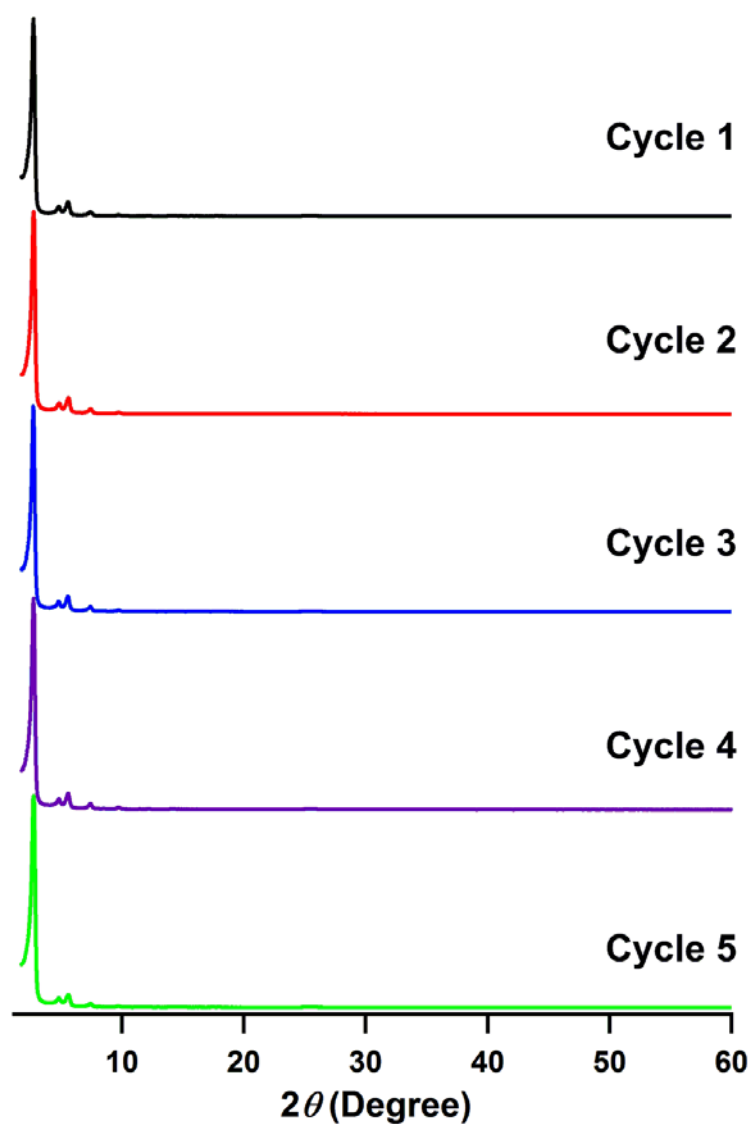
Supplementary Figure 20 | Nitrogen sorption isotherm profiles of $[S\text{-Py}]_{0.17}\text{-TPB-DMTP-COFs}$ after treated in different solvents for one week (black: as-synthesised, red: 12 M HCl, blue: boiling water, purple: 14 M NaOH). The filled circles represent adsorption and the open circles represent desorption.



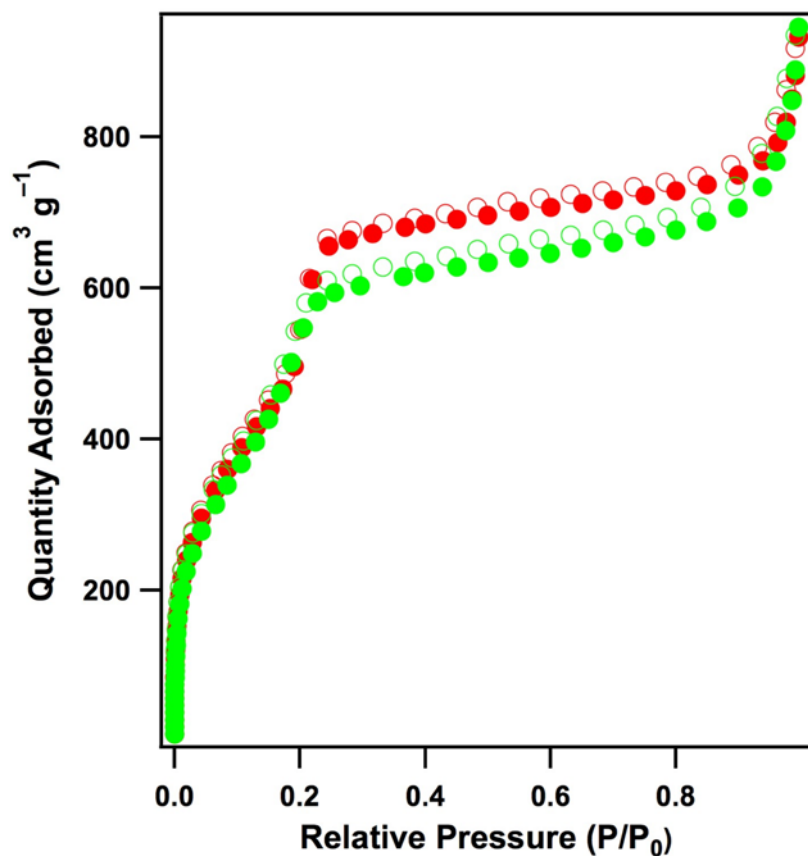
Supplementary Figure 21 | Conversion of the reactant (*trans*-2-chloro- β -nitrostyrene, red dots) and concentration of the corresponding product (blue dots) in the reaction system, determined by ^1H NMR.



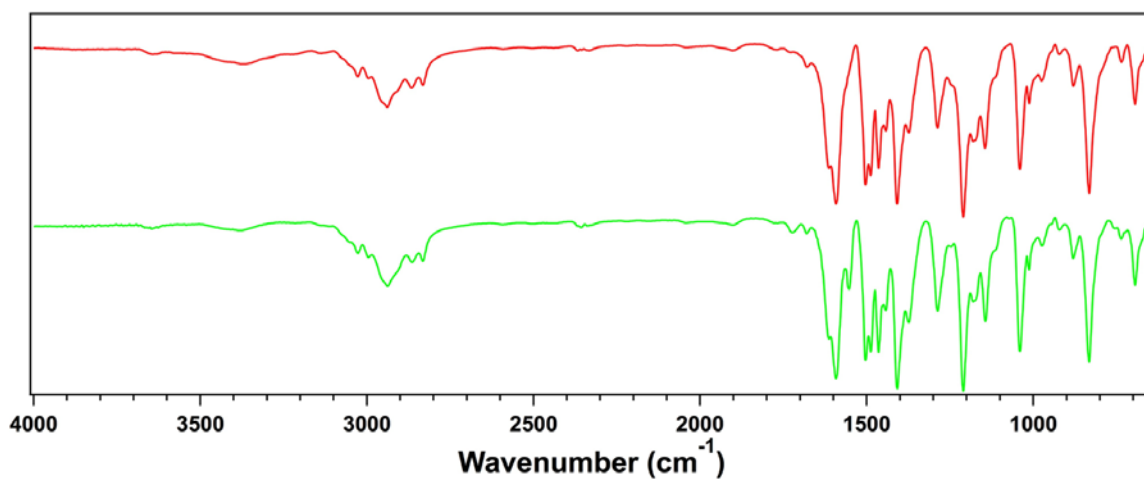
Supplementary Figure 22 | Natural logarithm plot of conversion of *trans*-2-chloro- β -nitrostyrene versus reaction time.



Supplementary Figure 23 | PXRD patterns of [S-Py]_{0.17}-TPB-DMTP-COF after each cycle.



Supplementary Figure 24 | Nitrogen sorption isotherm profiles of [S-Py]_{0.17}-TPB-DMTP-COF (red: fresh, green: after 5 cycles). The filled circles represent adsorption and the open circles represent desorption.



Supplementary Figure 25 | FT-IR spectra of $[S\text{-Py}]_{0.17}\text{-TPB-DMTP-COF}$ (red: fresh, green: after 5 cycles).

Supplementary Methods

Thin-layer chromatography (TLC) plates were visualized by exposure to ultraviolet light and/or developed with iodine vapor. Flash column chromatography was carried out with silica gel (200–300 mesh). ^1H nuclear magnetic resonance (NMR) spectra were recorded on JEOL models JNM-LA400 NMR spectrometers, where chemical shifts (δ in ppm) were determined with a residual proton of the solvent as standard. Fourier transform infrared (FT-IR) spectra were recorded on a JASCO model FT-IR-6100 infrared spectrometer. Powder X-ray diffraction (PXRD) data were recorded on a Rigaku model RINT Ultima III diffractometer by depositing powder on glass substrate, from $2\theta = 1.5^\circ$ up to 60° with 0.02° increment. Nitrogen sorption isotherms were measured at 77 K with a Micromeritics Instrument Corporation model 3Flex surface characterization analyzer. The Brunauer-Emmett-Teller (BET) method was utilized to calculate the specific surface areas. By using the non-local density functional theory (NLDFT) model, the pore volume was derived from the sorption curve. Elemental analysis was performed on a Yanako model CHN CORDER MT-6 elemental analyzer. High performance liquid chromatography was performed on a JASCO model HPLC model with Daicel chiral AD-H columns with *i*-PrOH/*n*-hexane as the eluent. Inductively coupled plasma atomic emission spectroscopy (ICP-AES) was conducted a Profile plus model of Teledyne Instruments Leeman Labs Inc. TGA measurements were performed on a Mettler-Toledo model TGA/SDTA851e under N_2 or O_2 , by heating to 800°C at a rate of $10^\circ\text{C min}^{-1}$. High-resolution transmission electron microscopy (HR-TEM) images were obtained on a JEOL model JEM-3200 microscopy. The sample was prepared by drop-casting a supersonicated tetrahydrofuran suspension of the COFs onto a copper grid.

The crystalline structures of COF were determined using the density-functional tight-binding (DFTB+) method including Lennard-Jones (LJ) dispersion. The calculations were carried out with the DFTB+ program package version 1.2⁹. DFTB is an approximate density functional theory method based on the tight binding approach and utilizes an optimized minimal LCAO Slater-type all-valence basis set in combination with a two-center approximation for Hamiltonian matrix elements. The Coulombic interaction between partial atomic charges was determined using the self-consistent charge (SCC) formalism. Lennard-Jones type dispersion was employed in all calculations to describe van der Waals (vdW) and π -stacking interactions. The lattice dimensions were optimized simultaneously with the geometry. Standard DFTB parameters for X–Y element

pair (X , $Y = \text{C, O, H and N}$) interactions were employed from the mio-0-1 set¹⁰. The accessible surface areas were calculated from the Monte Carlo integration technique using a nitrogen-size probe molecule (diameter = 3.68 Å) roll over the framework surface with a grid interval of 0.25 Å¹¹.

The XRD pattern simulation was performed in a software package for crystal determination from PXRD pattern, implemented in MS modeling version 4.4 (Accelrys Inc.). We performed Pawley refinement to optimize the lattice parameters iteratively until the R_p and R_{wp} values converge. The pseudo-Voigt profile function was used for whole profile fitting and Berrar-Baldinozzi function was used for asymmetry correction during the refinement processes. The final R_p and R_{wp} values were 2.02% and 4.37%. We also performed Rietveld refinement to optimize the lattice parameters iteratively until the R_p and R_{wp} values converge. The pseudo-Voigt profile function was used for whole profile fitting and Berrar-Baldinozzi function was used for asymmetry correction during the refinement processes. The final R_p and R_{wp} values were 3.31% and 6.52%.

Dehydrated N,N -dimethylformide (DMF), N,N -dimethylacetamide (DMAc), dehydrated tetrahydrofuran (THF), and o -dichlorobenzene (o -DCB) were purchased from Kanto Chemicals. Hydrobromic acid, p -toluenesulonyl chloride, trifluoroacetic acid, toluene, dioxane, mesitylene, 1-butanol, ethanol, and acetic acid were purchased from Wako Chemicals. Propargyl bromide, 1,4-dimethoxybenzene, N -(*tert*-butoxycarbonyl)-L-prolinol, *tert*-butyl alcohol, 1,3,5-tri(4-aminophenyl) benzene and benzoic acid were purchased from TCI. *trans*- β -Nitrostyrene, N,N -diisopropylethylamine, *trans*-4-chloro- β -nitrostyrene, *trans*-2-chloro- β -nitrostyrene, *trans*-4-bromo- β -nitrostyrene, *trans*-4-methyl- β -nitrostyrene, and *trans*-4-methoxy- β -nitrostyrene were purchased from Sigma-Aldrich Co.

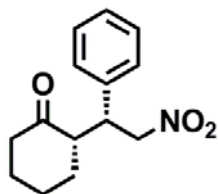
2,5-Dimethoxyterephthalaldehyde (DMTA), 2,5-dihydroxyterephthalaldehyde (DHTA) and 2,5-bis(2-propynyloxy) terephthalaldehyde (BPTA) were synthesized according to the reported methods^{12,13}. (*S*)-2-(Azidomethyl)pyrrolidine was synthesized using a reported method¹⁴.

TPB-TP-COF. An o -DCB/BuOH (0.5 mL/0.5 mL) mixture of TAPB (0.080 mmol, 28.1 mg) and TA (0.120 mmol, 16.1 mg) in the presence of acetic acid catalyst (6 M, 0.1 mL) in a Pyrex tube (10 mL) was degassed through three freeze–pump–thaw cycles. The tube was sealed by flame and heated at 120 °C for 3 days. The precipitate was collected *via* centrifugation, washed 6 times with THF, acetonitrile, and then subjected to Soxhlet extraction with THF as the solvent for 1 day to

remove trapped guest molecules. The powder was collected and dried at 120 °C under vacuum overnight to produce TPB-TP-COF in an isolated yield of 83%.

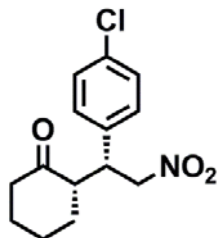
TPB-DHTP-COF. An *o*-DCB/BuOH (0.5 mL/0.5 mL) mixture of TAPB (0.080 mmol, 28.1 mg) and DHTA (0.120 mmol, 19.9 mg) in the presence of acetic acid catalyst (6 M, 0.1 mL) in a Pyrex tube (10 mL) was degassed through three freeze–pump–thaw cycles. The tube was sealed by flame and heated at 120 °C for 3 days. The precipitate was collected *via* centrifugation, washed 6 times with THF, acetonitrile and then subjected to Soxhlet extraction with THF as the solvent for 1 day to remove trapped guest molecules. The powder was collected and dried at 120 °C under vacuum overnight to produce TPB-DHTP-COF in an isolated yield of 81%.

(*S*)-2-((*R*)-2-Nitro-1-phenylethyl)cyclohexanone¹⁴⁻¹⁶



Major diastereomer: ¹H NMR (400 MHz, CDCl₃): δ = 7.34–7.25 (m, 3H), 7.15 (d, *J* = 7.2 Hz, 2H), 4.93 (dd, *J* = 12.4, 4.4 Hz, 1H), 4.62 (dd, *J* = 12.4, 10 Hz, 1H), 3.75 (dt, *J* = 10, 4.4 Hz, 1H), 2.72–2.64 (m, 1H), 2.51–2.44 (m, 1H), 2.42–2.33 (m, 1H), 2.11–2.03 (m, 1H), 1.81–1.66 (m, 4H), 1.28–1.16 (m, 1H). HPLC conditions: The enantiomeric excess was determined by HPLC (Chiralcel AD-H), hexane : *i*-PrOH = 90 : 10, UV = 206 nm, 25 °C, 0.5 mL min^{−1}, syn: *t*_R = 24.2 min (minor) and *t*_R = 29.8 min (major).

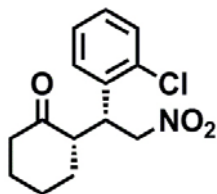
(*S*)-2-((*R*)-1-(4-Chlorophenyl)-2-nitroethyl)cyclohexanone¹⁷



Major diastereomer: ¹H NMR (400 MHz, CDCl₃): δ = 7.29 (d, *J* = 8.4 Hz, 2H), 7.10 (d, *J* = 8.4 Hz, 2H), 4.92 (dd, *J* = 12.4, 4.8 Hz, 1H), 4.59 (dd, *J* = 12.4, 9.6 Hz, 1H), 3.74 (dt, *J* = 9.8, 4.8 Hz, 1H), 2.68–2.59 (m, 1H), 2.50–2.43 (m, 1H), 2.41–2.31 (m, 1H), 2.12–2.04 (m, 1H), 1.83–1.76 (m, 1H), 1.75–1.53 (m, 3H), 1.28–1.16 (m, 1H). HPLC conditions: The enantiomeric excess was determined by HPLC (Chiralcel AD-H), hexane : *i*-PrOH = 90 : 10, UV = 206 nm, 20 °C, 0.5 mL min^{−1}, syn: *t*_R

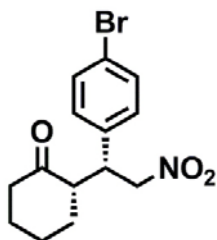
= 34.6 min (minor) and t_R = 50.5 min (major).

(*S*)-2-((*R*)-1-(2-Chlorophenyl)-2-nitroethyl)cyclohexanone¹⁷



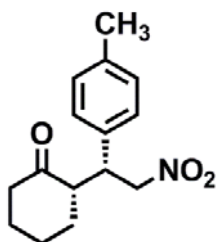
Major diastereomer: ¹H NMR (400 MHz, CDCl₃): δ = 7.37 (dd, J = 8.4, 1.6 Hz, 1H), 7.26-7.17 (m, 3H), 4.92-4.86 (m, 2H), 4.31-4.23 (m, 1H), 2.96-2.85 (m, 1H), 2.50-2.43 (m, 1H), 2.42-2.33 (m, 1H), 2.13-2.05 (m, 1H), 1.85-1.56 (m, 4H), 1.38-1.28 (m, 1H). HPLC conditions: The enantiomeric excess was determined by HPLC (Chiralcel AD-H), hexane : *i*-PrOH = 90 : 10, UV = 206 nm, 20°C, 0.5 mL min⁻¹, syn: t_R = 24.0 min (minor) and t_R = 39.2 min (major).

(*S*)-2-((*R*)-1-(4-Bromophenyl)-2-nitroethyl)cyclohexanone¹⁸



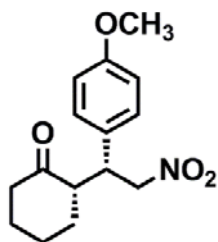
Major diastereomer: ¹H NMR (400 MHz, CDCl₃): δ = 7.44 (d, J = 8.4 Hz, 2H), 7.05 (d, J = 8 Hz, 2H), 4.92 (dd, J = 12.8, 4.8 Hz, 1H), 4.59 (dd, J = 12.8, 10 Hz, 1H), 3.73 (dt, J = 10, 4.4 Hz, 1H), 2.67-2.59 (m, 1H), 2.50-2.43 (m, 1H), 2.41-2.32 (m, 1H), 2.12-2.05 (m, 1H), 1.83-1.56 (m, 4H), 1.28-1.17 (m, 1H). HPLC conditions: The enantiomeric excess was determined by HPLC (Chiralcel AD-H), hexane : *i*-PrOH = 90 : 10, UV = 208 nm, 25°C, 1.0 mL min⁻¹, syn: t_R = 16.4 min (minor) and t_R = 24.7 min (major).

(*S*)-2-((*R*)-1-(4-Methylphenyl)-2-nitroethyl)cyclohexanone¹⁷



Major diastereomer: ^1H NMR (400 MHz, CDCl_3): δ = 7.11 (d, J = 8 Hz, 2H), 7.03 (d, J = 8.4 Hz, 2H), 4.90 (dd, J = 12.4, 4.8 Hz, 1H), 4.59 (dd, J = 12, 9.6 Hz, 1H), 3.70 (dt, J = 10, 4.8 Hz, 1H), 2.69-2.61 (m, 1H), 2.50-2.43 (m, 1H), 2.41-2.32 (m, 1H), 2.30 (s, 3H), 2.10-2.02 (m, 1H), 1.81-1.56 (m, 4H), 1.28-1.18 (m, 1H). HPLC conditions: The enantiomeric excess was determined by HPLC (Chiralcel AD-H), hexane : *i*-PrOH = 90 : 10, UV = 206 nm, 20°C, 0.5 mL min $^{-1}$, syn: t_R = 22.0 min (minor) and t_R = 28.2 min (major).

(*S*)-2-((*R*)-1-(4-Methoxyphenyl)-2-nitroethyl)cyclohexanone^{17,19}



Major diastereomer: ^1H NMR (400 MHz, CDCl_3): δ = 7.06 (d, J = 8.8 Hz, 2H), 6.83 (d, J = 9.2 Hz, 2H), 4.89 (dd, J = 12.4, 4.8 Hz, 1H), 4.57 (dd, J = 12.8, 10 Hz, 1H), 3.77 (s, 3H), 3.70 (dt, J = 10, 4.4 Hz, 1H), 2.67-2.59 (m, 1H), 2.49-2.43 (m, 1H), 2.41-2.32 (m, 1H), 2.10-2.02 (m, 1H), 1.81-1.52 (m, 4H), 1.28-1.18 (m, 1H). HPLC conditions: The enantiomeric excess was determined by HPLC (Chiralcel AD-H), hexane : *i*-PrOH = 80 : 20, UV = 206 nm, 20°C, 0.5 mL min $^{-1}$, syn: t_R = 21.7 min (minor) and t_R = 26.0 min (major).

Supplementary References

1. Marigo, M., Franzen, J., Poulsen, T. B., Zhuang, W. & Jorgensen, K. A. Asymmetric organocatalytic epoxidation of α,β -unsaturated aldehydes with hydrogen peroxide. *J. Am. Chem. Soc.* **127**, 6964-6965 (2005).
2. Hayashi, Y., Gotoh, H., Hayashi, T. & Shoji, M. Diphenylprolinol silyl ethers as efficient organocatalysts for the asymmetric Michael reaction of aldehydes and nitroalkenes. *Angew. Chem. Int. Ed.* **44**, 4212-4215 (2005).
3. Jensen, K. L., Dickmeiss, G., Jiang, H., Albrecht, L. & Jorgensen, K. A. The diarylprolinol silyl ether system: a general organocatalyst. *Acc. Chem. Res.* **45**, 248-264 (2012).
4. Kresge, C. T., Leonowicz, M. E., Roth, W. J., Vartuli, J. C. & Beck, J. S. Ordered mesoporous molecular sieves synthesized by a liquid-crystal template mechanism. *Nature* **359**, 710-712 (1992).
5. Férey, G. *et al.* A chromium terephthalate-based solid with unusually large pore volumes and surface area. *Science* **309**, 2040-2042 (2005).
6. Nagai, A. *et al.* Pore surface engineering in covalent organic frameworks. *Nature Commun.* **2**, 536 (2011).
7. Xu, H. *et al.* Catalytic covalent organic frameworks via pore surface engineering. *Chem. Commun.* **50**, 1292-1294 (2014).
8. Chen, L. *et al.* Photoelectric covalent organic frameworks: converting open lattices into ordered donor-acceptor heterojunctions. *J. Am. Chem. Soc.* **136**, 9806-9809 (2014).
9. Aradi, B., Hourahine, B. & Frauenheim, T. DFTB+, a sparse matrix-based implementation of the DFTB method. *J. Phys. Chem. A* **111**, 5678-5684 (2007).
10. Lukose, B., Kuc, A. & Heine, T. Stability and electronic properties of 3D covalent organic frameworks. *J. Mol. Model.* **19**, 2143-2148 (2013).
11. Dören, T., Millange, F., Férey, G., Walton, K. S. & Snurr, R. Q. Calculating Geometric Surface Areas as a Characterization Tool for Metal–Organic Frameworks. *J. Phys. Chem. C* **111**, 15350-15356 (2007).
12. Palmgren, A., Thorarensen, A. & Backvall, J. E. Efficient synthesis of symmetrical 2,5-disubstituted benzoquinones via palladium-catalyzed double Negishi coupling. *J. Org. Chem.* **63**, 3764-3768 (1998).

13. Lingam, V. S. P. R., Vinodkumar, R., Mukkanti, K., Thomas, A. & Gopalan, B. A simple approach to highly functionalized benzo[b]furans from phenols and aryl iodides via aryl propargyl ethers. *Tetrahedron Lett.* **49**, 4260-4264 (2008).
14. Luo, S. *et al.* Evolution of pyrrolidine-type asymmetric organocatalysts by "click" chemistry. *J. Org. Chem.* **71**, 9244-9247 (2006).
15. Pansare, S. V. & Pandya, K. Simple diamine- and triamine-protonic acid catalysts for the enantioselective Michael addition of cyclic ketones to nitroalkenes. *J. Am. Chem. Soc.* **128**, 9624-9625 (2006).
16. Cobb, A. J., Shaw, D. M., Longbottom, D. A., Gold, J. B. & Ley, S. V. Organocatalysis with proline derivatives: improved catalysts for the asymmetric Mannich, nitro-Michael and aldol reactions. *Org. Biomol. Chem.* **3**, 84-96 (2005).
17. Luo, S. *et al.* Functionalized chiral ionic liquids as highly efficient asymmetric organocatalysts for Michael addition to nitroolefins. *Angew. Chem. Int. Ed.* **45**, 3093-3097 (2006).
18. Cao, C. L., Ye, M. C., Sun, X. L. & Tang, Y. Pyrrolidine-thiourea as a bifunctional organocatalyst: highly enantioselective Michael addition of cyclohexanone to nitroolefins. *Org. Lett.* **8**, 2901-2904 (2006).
19. Ishii, T., Fujioka, S., Sekiguchi, Y. & Kotsuki, H. A new class of chiral pyrrolidine-pyridine conjugate base catalysts for use in asymmetric Michael addition reactions. *J. Am. Chem. Soc.* **126**, 9558-9559 (2004).